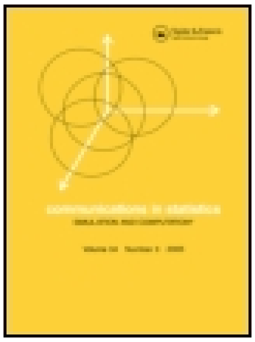


Communications in Statistics - Simulation and Computation

ISSN: 0361-0918 (Print) 1532-4141 (Online) Journal homepage: http://www.tandfonline.com/loi/lssp20

An Evaluation of Change-Point Estimators for a Sequence of Normal Observations with Unknown Parameters

Jorge Garza-Venegas, Victor Tercero-Gomez, Alvaro Cordero Franco, María Temblador-Pérez & Mario Beruvides

An Evaluation of Change-Point Estimators for a Sequence of Normal Observations with Unknown Parameters

Garza-Venegas, Jorge, Tercero-Gomez, Victor*, Cordero Franco, Alvaro, Temblador-Pérez, María, Beruvides, Mario

Corresponding Author Email: victor.tercero@itesm.mx

Performance of maximum likelihood estimators (MLE) of the change-point in normal series is evaluated considering three scenarios where process parameters are assumed to be unknown. Different shifts, sample sizes, and locations of a change-point were tested. A comparison is made with estimators based on cumulative sums and Bartlett's test. Performance analysis done with extensive simulations for normally distributed series showed that the MLEs perform better (or equal) in almost every scenario, with smaller bias and standard error. In addition, robustness of MLE to non-normality is also studied.

Keywords:

MLE; CUSUM; Bartlett's Test; unknown parameters; robustness

1. Introduction

Change-point analysis could be thought as a collection of tools or techniques to get a response to Arunajadai's (2009, p.58) question: ―Are the data homogeneous and if not, what are the locations of the homogeneous segments in the data?" The answer might be viewed as a key element within the field of Statistical Process Control (SPC), where the knowledge of a change is linked to the concept of assignable causes of variation. By addressing these causes, practitioners might boost the application of corrective actions, and, therefore, reduce quality costs (inspection, rework, scrap, downtime, defects, etc.) of products and processes. That is, knowledge of a change-point location, when it exists, facilitates process improvement to secure an in control state.

Since its formulation, the change-point problem has been addressed from different approaches such as parametric, nonparametric, Bayesian, among others, and considering different factors such as sample size or series length, providing limit theorems and asymptotic distribution of the change-point (see, for instance, Fotopoulos et al. [2010]); or considering phases of SPC. A Phase I requires the estimation of parameters from the analysis of historical data; and during a Phase II, a process initiates an online monitoring.

From a parametric point of view, following the guidelines of Hinkley (1970), several authors like Samuel et al. (1998a, 1998b) and Khoo (2004) have studied the behaviour of the maximum likelihood estimators (MLE) for a change-point in series of normal observations (see Chen and Gupta [2001] for an in deep review of the parametric problem). These authors studied the behaviour of these estimators where a priori knowledge of initial parameters existed. Nevertheless, real life applications usually lack this information, and parameters have to be estimated from collected data. Moreover, limited sample size (or series length) has not been deeply studied in the literature, only limit theorems and asymptotic distributions are given, leaving a gap in the corresponding knowledge. An in-depth study of biases and standard errors of change-point estimators applied in retrospective analysis was not found, and it is believed that the corresponding performance analysis can be used to determine practical application strategies and generate confidence in the methods when used in real life situations.

This paper is focused on the performance evaluation of several change-point MLEs for normal observations collected over a discrete time where parameters before and after the change-point are not known. Thus, given a time series of independent observations $X_{1,1},\ldots,X_{1,n},\ldots,X_{\tau,1},\ldots,X_{\tau,n},X_{\tau+1,1},\ldots,X_{\tau+1,n},\ldots,X_{\mathrm{T},1},\ldots,X_{\mathrm{T},n}$ following the subsequent model:

$$X_{i,j} \sim \begin{cases} N\left(\mu_0,\sigma_0\right), & 1 \le i \le \tau, 1 \le j \le n \\ N\left(\mu_1,\sigma_1\right), & \tau < i \le \mathrm{T}, 1 \le j \le n \end{cases} \tag{1}$$

where parameters τ, μ_0, μ_1, σ_0 and σ_1 are unknown, and τ is called the change-point of the series, where T is the last sample observed. Change-point MLEs are obtained based on three different scenarios according to different assumptions about parameters of the normal distribution:

1. $\hat{\tau}_{MLE1}$ assumes that $\mu_0 \ne \mu_1$ but $\sigma_0 = \sigma_1$.
2. $\hat{\tau}_{MLE2}$ assumes that $\sigma_0 \ne \sigma_1$ but $\mu_0 = \mu_1$.
3. $\hat{\tau}_{MLE3}$ assumes $\mu_0 \ne \mu_1$ and $\sigma_0 \ne \sigma_1$.

Extensive Monte Carlo simulations are used to measure the performance of these estimators. 10,000 replicates were used to assess different scenarios where the size of the shift, sample size, and the change-point position were modified to evaluate the sensitivity of estimators. Results are compared with known estimators from Hinkley (1971), for change in mean, and Hawkins and Zamba (2005), for change in variance.

There are two important facts to be mentioned about the scope of this study and its contribution. First one is that because estimators always give an estimation of the change-point whether or not it has already occurred, their use is recommended when assumption of knowledge of the change-point occur is met, which could be done with control charts or hypothesis tests. For example, Csörgö and Horváth (1997) and Chen and Gupta (2011) obtained limiting results and approximations, and Tercero et al. (2013) provided critical values of the likelihood ratio test (LRT) for a change in the mean over series up to 100. The second one, is that even though Hinkley (1971) provided a comparison between change-point MLE and CUSUM estimators as well as asymptotic properties of them, this study might be seen as an extension of his empirical results, since a wide variety of scenarios are considered here and a comparison against another estimators is presented. Moreover, his simulations only considered initial parameters as known.

The organization of this paper is as follows: Section 2 presents an overview of the research on change-point analysis. Section 3 presents the change-point estimators under analysis. Experimental results from Monte Carlo simulations are described in Section 4; and finally, some general conclusions and opportunities for future research are mentioned in Section 5.

2. Background. Previous Work on Change-Point Analysis

Concerns about detecting sustained changes in time series started with Girshick and Rubin (1952) from a Bayesian paradigm. They defined a quality control rule to trigger corrective actions when a change in a random process was detected. Soon after, Page (1954, 1955, 1957), from a frequentist point of view, developed the cumulative sum (CUSUM) chart to be able to detect, faster than traditional Shewhart (1931) control charts, the presence of sustained changes.

Later, Hinkley (1970) consolidates the theoretical foundation to construct MLEs and likelihood ratio (LR) tests to detect change points in series of independent random variables, and applied these techniques to series with normally distributed observations. Moreover, provides some asymptotic properties of this estimator (consistency) and of its distribution (symmetric) under some special cases. Following this theory, Samuel et al. (1998a, 1998b), Samuel and Pignatiello (1998), and Dabye and Kutoyants (2001) developed MLEs for series following different distributions. They focused their work on the integration with control charts, where the MLEs were used as a plug in to estimate the change already detected.

Also, Hawkins and Zamba (2005) presented a model to deal with variance changes, analogous to the generalized likelihood ratio (GLR) control chart for variance, based on Bartlett's test. Nevertheless, the proposed model grew in complexity as more observations were added into the time series. To solve the complexity issue of the GLR when dealing with mean changes, Reynolds and Jianying (2010) simplified the approach by suggesting the application of a moving window, where the amount of steps required with each new observation was restricted to the size of the window. These methods deal with online monitoring while off-line estimation (retrospective analysis) will be considered in this paper. For a review on retrospective analysis, the reader is referred to Gurevich and Vexler (2010).

Aside from MLE procedures, Page (1954), when developing the CUSUM charts, first pointed out that results from this technique could be used to estimate the initial time of a change. Then, Hinkley (1971) compared the later estimation with the MLE for known initial parameters of normal and independent observations, concluding that the use of the CUSUM approach is asymptotically biased, but easier to use. Furthermore, he proposed to use the overall sample mean in case of unknown parameters in order to use the CUSUM procedure, and some insights in order to develop the asymptotic theory of this estimator.

Other approaches to the change-point problem follow nonparametric and Bayesian philosophies. Bayesian works were initiated with Girshick and Rubin's (1952) work while Bhattacharya and Johnson (1968) were responsible for the first construction of a nonparametric test with the explicit idea of developing a nonparametric methodology. An interested reader is referred to Bhattacharya (1994) and Wolfe and Schechtman (1984) respectively, for a review of these techniques.

As can be seen, a detailed comparison between change-point MLEs and other estimators is needed. The following section introduces the MLEs for unknown parameters under three different circumstances when mean and variance are assumed to be equal or different. Hinkley (1971) and Hawkins and Zamba (2005) estimators are also presented.

3. Change-Point Estimators

Following the theory developed by Hinkley (1970), three different scenarios are studied, giving three different estimates: for changes in mean (Section 3.1), changes in variance (Section 3.2), and changes in mean and variance at the same time (Section 3.3). Hinkley (1971) and Hawkins and Zamba (2005) estimators are reviewed in Sections 3.4 and 3.5, respectively.

To derive the MLE, let $\{ x_{ij} : i = 1,2,...,\tau,\ j = 1,2,...,n \}$ be independent and identically distributed random variables (i.i.d.r.v.) $N(\mu_0, \sigma_0)$, and $\{ x_{ij} : i = \tau+1,\ \tau+2,...,T,\ j = 1,2,...,n \}$ be i.i.d.r.v. $N(\mu_1, \sigma_1)$. For a given value of τ, the likelihood function is:

$$L(\mu_0, \mu_1, \sigma_0, \sigma_1 | \vec{x}) = \prod_{i=1}^{\tau}\prod_{j=1}^{n} f(x_{ij} | \mu_0, \sigma_0) \cdot \prod_{i=\tau+1}^{T}\prod_{j=1}^{n} f(x_{ij} | \mu_1, \sigma_1) \tag{2}$$

and MLEs are found by finding the parameter values that maximizes the function in equation (2). Results are shown in the following three sections.

3.1 MLE when $\mu_0 \neq \mu_1$ and $\sigma_0 = \sigma_1$

Equations (3) through (5) show the corresponding MLEs for the parameters of a normal distribution. Equation (6) gives the change-point MLE of the model shown in equation (1) with corresponding assumptions of this scenario.

$$\hat{\mu}_0 = \frac{\sum_{i=1}^{\tau}\sum_{j=1}^{n} x_{ij}}{tn} \tag{3}$$

$$\hat{\mu}_1 = \frac{\sum_{i=\tau+1}^{T}\sum_{j=1}^{n} x_{ij}}{(\mathrm{T} - t)n} \tag{4}$$

$$\hat{\sigma}_p^{\ 2}(\hat{\mu}_0, \hat{\mu}_1) = \frac{\sum_{i=1}^{\tau}\sum_{j=1}^{n} (x_{ij} - \hat{\mu}_0)^2 + \sum_{i=\tau+1}^{T}\sum_{j=1}^{n} (x_{ij} - \hat{\mu}_1)^2}{n\mathrm{T}} \tag{5}$$

$$\hat{\tau}_{MLE1} = \underset{5 \le \tau \le \mathrm{T}-5}{\arg\min} \{ \hat{\sigma}_p (\hat{\mu}_0, \hat{\mu}_1) \} \tag{6}$$

Note that change-point MLEs omit the first and last five samples of the series. This is done to avoid the undesirably tendency of these MLEs to favour the locations of change-point at the beginning or end of the series, as suggested by Karl and Williams (1987). This was also suggested in Jann (2000) who estimated multiple change-points in normal series using an iterative method based on the t-statistic. For instance, for a series length of 50, a change in mean of 0.5 standard deviations, a change in the standard deviation ratio (σ_1/σ_0) equal to 1.1, by evaluating the likelihood function from 2 to T-2 = 48, with a change at moment $\tau / \mathrm{T} = 0.5$, and a subgroup size of n = 1, it is obtained a bias of $E(\hat{\tau} - \tau)$ = 1.26 and a standard error of $\sqrt{Var(\hat{\tau})}$ = 16.16. While a search between 5 to T-5 = 45 gives values of 0.58 and 11.72, respectively. This situation is repeated in every scenario studied. Hence, for the sake of brevity, further results are omitted as they only support what is already known. The assumption of not having a change-point within the first and last five samples is used when constructing the other change-point MLEs in this section.

3.2 MLE when $\mu_0 = \mu_1$ and $\sigma_0 \neq \sigma_1$

Equations (7) and (8) show the corresponding MLEs for the variances (before and after the change). The estimator of μ, $\hat{\mu}_p$, is obtained after solving equation (9), which is a third degree polynomial equation with at least one real solution out of three possible ones, which need to be evaluated individually using equation (10). The latter equation gives the MLE change-point of the model from equation (1) with the conditions of this second scenario.

$$\hat{\sigma}_0^2(\hat{\mu}_p) = \frac{\sum_{i=1}^{\tau}\sum_{j=1}^{n}\left(x_{ij} - \hat{\mu}_p\right)^2}{n\tau} \tag{7}$$

$$\hat{\sigma}_1^2(\hat{\mu}_p) = \frac{\sum_{i=\tau+1}^{T}\sum_{j=1}^{n}\left(x_{ij} - \hat{\mu}_p\right)^2}{n(\mathrm{T} - \tau)} \tag{8}$$

$$\frac{1}{\hat{\sigma}_0^2(\mu)}\sum_{i=1}^{\tau}\sum_{j=1}^{n}\left(x_{ij}-\mu\right)+\frac{1}{\hat{\sigma}_1^2(\mu)}\sum_{i=\tau+1}^{\mathrm{T}}\sum_{j=1}^{n}\left(x_{ij}-\mu\right)=0 \tag{9}$$

$$\hat{\tau}_{MLE2}=\underset{5\le\tau\le\mathrm{T}-5}{\arg\min}\left\{\hat{\sigma}_0^{\tau}\left(\hat{\mu}_p\right)\hat{\sigma}_1^{\mathrm{T}-\tau}\left(\hat{\mu}_p\right)\right\} \tag{10}$$

It can be shown that the Hessian Matrix of σ_0^2 and σ_1^2 is defined negative, independently of the value of μ (see proof in Tercero [2011], when $n = 1$). Nevertheless, equation (9) has three different roots (including imaginary ones), and an evaluation of the loglikelihood function is required to choose the maximum out of the real results. Equation (11) shows the corresponding loglikelihood function without the constant values. Given any t, it can be seen, in equation (11), that as $\mu\to\infty$ or $\mu\to-\infty$ then $\log L<0$, hence, the MLE for μ exists, and it can be found by evaluating the roots obtained in equation (9) in equation (11).

$$\begin{aligned}\log L=-n\tau\log\left(\hat{\sigma}_0\left(\hat{\mu}_p\right)\right)-n(\mathrm{T}-\tau)\log\left(\hat{\sigma}_1\left(\hat{\mu}_p\right)\right)-\frac{1}{2\hat{\sigma}_0^2\left(\hat{\mu}_p\right)}\sum_{i=1}^{\tau}\sum_{j=1}^{n}\left(x_{ij}-\hat{\mu}_p\right)^2\\-\frac{1}{2\hat{\sigma}_1^2\left(\hat{\mu}_p\right)}\sum_{i=\tau+1}^{\mathrm{T}}\sum_{j=1}^{n}\left(x_{ij}-\hat{\mu}_p\right)^2\end{aligned} \tag{11}$$

3.3 MLE with $\mu_0\neq\mu_1$ and $\sigma_0\neq\sigma_1$

Equations (7) and (8) are used to calculate the variances $\hat{\sigma}_0^2(\hat{\mu}_0)$ and $\hat{\sigma}_1^2(\hat{\mu}_1)$, where $\hat{\mu}_0$ and $\hat{\mu}_1$ are used instead of $\hat{\mu}_p$. Equation (12) gives the MLE change-point of the model from equation (1) with the conditions of this third scenario.

$$\hat{\tau}_{MLE3}=\underset{5\le\tau\le\mathrm{T}-5}{\arg\min}\left\{\hat{\sigma}_0^{\tau}\left(\hat{\mu}_0\right)\hat{\sigma}_1^{\mathrm{T}-\tau}\left(\hat{\mu}_1\right)\right\} \tag{12}$$

If variance values are smaller than 1, the minimization process over equations: (6), (10) and (12) can be made over the logarithms of these functions to reduce the risk of rounding errors.

3.4 Change-Point estimation with CUSUM for shifts in mean

From Hinkley (1971, p.517), estimation of the change-point for mean shifts, with constant variance, is obtained by finding the maximum absolute cumulated error from the overall mean. Deviation is calculated according to equation (13) and the change-point estimator is given in (14).

$$S_0 = 0,\ S_t = \sum_{i=1}^{t}\left(X_i - \mathrm{T}^{-1}\sum_{i=1}^{\mathrm{T}} X_i \right),\ t = 1,2,\ldots,T \tag{13}$$

$$\hat{\tau}_{CUSUM} = \inf_{t_0}\left\{ t_0 : \left|S_{t_0}\right| \geq \left|S_t\right|; t = 1,\ldots,\mathrm{T} \right\} \tag{14}$$

If subgroups exist, mean values within each subgroup of X_i observations are used as if they were individual observations. It has to be noted that this estimator is proposed when parameters are unknown and it was adapted to address changes in an unknown direction.

3.5 Change-Point estimation with Bartlett's test statistic

To assess for variance changes, Hawkins and Zamba (2005) used the Bartlett's test statistic and the GLR approach to create a change-point estimator. The change-point estimator is obtained by finding the moment t where equation (15), Bartlett's test statistic, is maximized. This equation uses equations (3), (5), (7), and (8) to obtain values for $\hat{\mu}_0$, $\hat{\mu}_1$, $\hat{\sigma}_0^2(\hat{\mu}_0)$, $\hat{\sigma}_1^2(\hat{\mu}_1)$ and $\hat{\sigma}_p^2(\hat{\mu}_0, \hat{\mu}_1)$.

$$G_t = \frac{(nt-1)\log\left[k_1 \dfrac{\hat{\sigma}_p^2(\hat{\mu}_0, \hat{\mu}_1)}{\hat{\sigma}_0^2(\hat{\mu}_0)} \right] + \left(n(\mathrm{T}-t)-1\right)\log\left[k_2 \dfrac{\hat{\sigma}_p^2(\hat{\mu}_0, \hat{\mu}_1)}{\hat{\sigma}_1^2(\hat{\mu}_1)} \right]}{C} \tag{15}$$

where:

$$C = 1 + \left[(nt-1)^{-1} + \left(n(\mathrm{T}-t)-1\right)^{-1} - (n\mathrm{T}-2)^{-1} \right] / 3 \tag{16}$$

$$k_1 = \frac{T(nt-1)}{t(n\mathrm{T}-2)} \tag{17}$$

$$k_2 = \frac{T(n(T-t)-1)}{(T-t)(n\mathrm{T}-2)} \tag{18}$$

As a result, Hawkins and Zamba's (2005) change-point estimator can be written as:

$$\hat{\tau}_G = \underset{2 \le t \le \mathrm{T}-2}{\arg\max}\{G_t\} \tag{19}$$

4. Performance of Estimators

4.1 Retrospective Analysis: Simulation Design

To evaluate performance of $\hat{\tau}_{MLE1}$, $\hat{\tau}_{MLE2}$, $\hat{\tau}_{MLE3}$, $\hat{\tau}_{CUSUM}$ and $\hat{\tau}_G$, several scenarios were simulated by considering different shifts of the mean (δ), ratio of deviations (σ_1/σ_0), series length (T), subgroup sizes (n), and change-point position (τ/T). Monte Carlo experimentation was used to evaluate the performance of the change-point estimators in terms of bias, $E(\hat{\tau}-\tau)$, and standard error, $\sqrt{Var(\hat{\tau})}$. Simulations were made as follows: a scenario is selected (different values of shifts in the mean, ratio of deviations, etc.), after that, a series under that scenario is created, and estimators under analysis are calculated for that series. This is repeated 10,000 times and then bias and standard error of estimators are calculated. It is worth noting that some scenarios might look similar. For instance, scenarios with T = 100, n = 3, and T = 300, n = 1 might look the same, but a difference is expected in performance, since the change-point sample space is {5,6,…,95} in the former, and {5,6,…,295} in the latter.

4.2 Simulation Results

4.2.1. Normal assumption.

When $\mu_0 \neq \mu_1$ and $\sigma_0 = \sigma_1$, $\hat{\tau}_{MLE1}$ and $\hat{\tau}_{CUSUM}$ were evaluated over different series length (T), sample sizes (*n*), shifts in mean (δ) measured in standard deviations, and change-point locations (τ / T). Table 1 shows, when $\tau/\mathrm{T} = 0.5$, that $\hat{\tau}_{MLE1}$ and $\hat{\tau}_{CUSUM}$ perform with similar biases—with a difference between estimators of at most ± 0.1—but $\hat{\tau}_{CUSUM}$ presents a smaller standard error than $\hat{\tau}_{MLE1}$. In addition to estimations in Table 1, Figures 1 to 6 show bar graphs of relative frequencies of $\hat{\tau} - \tau$ for both estimators according to few scenarios (Figures 1 through 3 show scenarios with T = 50, δ = 1, τ / T = 0.5 and n=1, 3 and 5, respectively while Figures 4 to 6 scenarios considering T = 100, n = 1, τ / T = 0.5 and δ = 1, 1.5 and 3, respectively). These figures might imply that both estimators are the same, however, when $\tau/\mathrm{T} \neq 0.5$ (Table 2), $\hat{\tau}_{MLE1}$ have a smaller bias and standard error in almost all cases, as can be seen in Figure 7, (fixing T = 50, n = 1, τ / T = 0.2 and δ = 1.5). Considering that, in practice, an analyst is not assumed to know beforehand when the change actually occurred, then $\hat{\tau}_{MLE1}$ is the recommended estimator, because it is more robust to the change-point location.

Performance results of $\hat{\tau}_{MLE2}$ and $\hat{\tau}_G$ when $\mu_0 = \mu_1$ and $\sigma_0 \neq \sigma_1$ are shown in Tables 3 and 4. It can be seen, when $\tau/\mathrm{T} = 0.5$, that $\hat{\tau}_{MLE2}$ and $\hat{\tau}_G$ have almost the same performance. Biases and standard errors are within ± 1 in most cases (in both tables) . $\hat{\tau}_G$ tends to have a bigger bias but a smaller standard error, mostly for smaller samples. For instance, Figure 8 consider T = 50, n = 1, τ / T = 0.2 and σ_1 / σ_0 = 1.5. However, when samples are large enough $\hat{\tau}_{MLE2}$ and $\hat{\tau}_G$ perform alike. This might suggest that estimators are highly correlated. Even more, a reader might think that they are the same estimators, and the difference is due to the randomness of the Monte Carlo simulation. Nevertheless, estimators are, in fact different, which could be pointed out because estimators have been applied to the same series in every scenario, as was explained above. It seems that both estimators converge at the same values at some point. On the other hand $\hat{\tau}_G$ requires less operations to be obtained, hence it requires less computer time.

If it is not possible to assume that only one parameter changed at the change-point, one should assume $\mu_0 \neq \mu_1$ and $\sigma_0 \neq \sigma_1$. Tables 5 and 6 compare this situation using $\hat{\tau}_{MLE3}$ with $\hat{\tau}_{CUSUM}$ and $\hat{\tau}_G$, (described in equations 12, 14 and 19). In Table 5, using T = 30 and $\tau/\mathrm{T} = 0.5$, when comparing part (a) versus part (b), it can be seen that as the ratio of standard deviations increases, the error bias and standard error of the $\hat{\tau}_{CUSUM}$ also increase, while the $\hat{\tau}_{MLE3}$ improves its precision. Table 5 (c) shows that $\hat{\tau}_G$ performs relatively well when there is no change in the mean, only in variance, its bias is similar to the $\hat{\tau}_{MLE3}$ and the standard error is smaller. Nevertheless, when $\delta > 0$, bias and standard error of $\hat{\tau}_{MLE3}$ tend to be smaller than $\hat{\tau}_G$ in every case. Additional results, not presented in the paper, using T = 100 showed the same differences, but with more pronounced effects. Table 6 confirms the superiority of $\hat{\tau}_{MLE3}$ over $\hat{\tau}_{CUSUM}$ and $\hat{\tau}_G$ when changes in mean and variance occur at different τ/T. In almost every case, $\hat{\tau}_{MLE3}$ give a smaller bias and standard error (for instance, see Figure 9 with T = 50, n = 1, σ_1/σ_0 = 1.5, δ = 1.5, τ/T = 0.2). Table 6 also shows that performance of $\hat{\tau}_G$ is undesirable, because bias and standard error increases as the series length does. This is probably due to the fact that it was created for changes only in variance, so simulation results might imply $\hat{\tau}_G$ is sensitive to shifts in the mean.

Furthermore, in order to determine what happens when assuming a false alarm (when $\delta = 0$ and $\sigma_1/\sigma_0 = 1$), Table 7 show that estimations are around the center of the series for all $\hat{\tau}_{MLE1}$, $\hat{\tau}_{MLE2}$, $\hat{\tau}_{MLE3}$, $\hat{\tau}_{CUSUM}$ and $\hat{\tau}_G$ estimators but the standard deviation is quite large.

Finally, Tables 1 through 6 show that $\hat{\tau}_{MLE1}$, $\hat{\tau}_{MLE2}$, $\hat{\tau}_{MLE3}$, $\hat{\tau}_{CUSUM}$ and $\hat{\tau}_G$ are not consistent estimators with respect to the series length. Hinkley (1970) mentioned this situation for $\hat{\tau}_{MLE1}$. However, the bias and standard error of these estimators decreases as n is increased, so they are consistent with respect to n, as long as the corresponding assumptions for each estimator are met.

Overall, when comparing $\hat{\tau}_{MLE3}$ with $\hat{\tau}_{MLE1}$ and $\hat{\tau}_{MLE2}$ it can be seen that performance of $\hat{\tau}_{MLE3}$ is not bad (absolute bias is smaller than the unit in most cases, standard error is similar to the other MLE's, and it shows consistency as the series length increases). This makes $\hat{\tau}_{MLE3}$ a safe choice when compared with the other estimators that need to assume that only one parameter changed.

4.2.2 Robustness to non normality.

Away from normal distributions, t (with 3 degrees of freedom) and gamma (with location parameter $\alpha = 0.5$ and scale parameter $\beta = 1$) distributions were considered and performance of the estimators was evaluated and are shown in Tables 8 to 10. Gamma observations were centered to provide independent control over mean and variance changes during the simulations. Table 8 shows that performance of both $\hat{\tau}_{MLE1}$ and $\hat{\tau}_{CUSUM}$ is quite similar for both distributions, but, bias is bigger using gamma distribution. When $\mu_0 = \mu_1$ and $\sigma_0 \neq \sigma_1$, Table 9 indicates that, for both distributions (t and gamma), estimators $\hat{\tau}_{MLE2}$ and $\hat{\tau}_G$ are sensitive to change-point locations (small for $\tau/\mathrm{T} = 0.5$, and big for $\tau/\mathrm{T} = 0.2$), however, standard error is clearly large, even if it decreases as the series length increases, it does not decreases as fast as in the normal case.

Additionally, for change in both parameters and using $\hat{\tau}_{MLE3}$, Table 10 parts (a) and (d) show that even though biases are bigger than the normal distribution case, differences in biases are less than unity when using t distribution, which is not the case for the gamma distribution. Nevertheless, standard error is clearly bigger when using both distributions. For $\hat{\tau}_{CUSUM}$, results in Table 10 parts (b) and (e) show that biases for both t and gamma distributions are smaller than when using the normal distribution, but standard error is bigger in almost every case. Finally, Table 10 parts (c) and (f) show that $\hat{\tau}_G$ has bigger standard error in almost every case. As done in the normal case, additional results, not presented in the paper, using T = 100 the effects are pronounced, but conclusions were the same.

5. Conclusions and Future Work

This paper presented the maximum likelihood estimators of the change-point in a series of independent normal observations that deal with changes in the mean, the variance, and in both parameters at the same time, with no prior knowledge about parameters. Estimators were obtained by maximizing the likelihood function of time series when there is an unknown change at moment τ and the mean and variance before and after the change are also unknown. Some of these estimators can be found in the literature; however there was no indication about their performance with fixed samples and their relationship with variance minimization (it was shown that all estimators could be written as a variance minimization problem). Also, no reference was found concerning $\hat{\tau}_{MLE2}$ estimator, probably due to the lack of a close-form expression. However, it was shown that $\hat{\tau}_{MLE2}$ was in fact a MLE since it provides a global maximum for the likelihood function.

The MLEs here shown were evaluated over different shifts, sample sizes, and change-points locations; and they were compared with two estimators, one based on the cumulative sum (CUSUM) and the other on Bartlett's test. The former, the CUSUM estimator, was designed to detect the moment when a change occurred in the mean of normal observations, and the latter, when the change occurs in the variance. Performance was measured using extensive simulations, showing that the MLEs perform better (or equal) in almost every scenario when compared with their counterparts. Bias and standard error was generally smaller with the MLEs. Moreover, numerical results shows consistency of $\hat{\tau}_{MLE1}$ and $\hat{\tau}_{MLE3}$.Nonetheless, it is important to notice that CUSUM based estimators are easier to compute, and their performances could be acceptable in some situations, making them an alternative when only rough estimates are needed.

Some extra scenarios were simulated where the change-point location was chosen to be closer to the beginning of the series, but simulation results are not presented here for the sake of brevity and they only provide numerical evidence of another asymptotic result from Hinkley (1970): $\hat{\tau}_{MLE1}$ distribution is practically symmetrical (from asymptotic theory this is ensured when dealing with symmetrical distributions and change is in location). This holds unless the change-point is right at the border of the series. Results of the other estimators suggests that they are not symmetrical distributed.

Finally, the non-normal scenarios evaluated here suggest that most change-point MLEs are sensitive to changes in variance. Only $\hat{\tau}_{MLE1}$ performance was found to be robust to non-normality. Future work will address other types of situations seen in a Phase I or II of SPC. It is known that types of changes found in a sequence of Phase I process data are not necessarily that of a single shift. Instead, if the system is out-of-control, multiple change points might be at hand in the form of shifts in mean, variance, drifts of a trend, seasonality, outliers, or a combination of these over independent or autocorrelated observations. Future research should focus on the development of multiple change-points, autocorrelated data, and non-normal observations through nonparametric change-point estimators.

 ACCEPTED MANUSCRIPT

References

Arunajadai, S.G. (2009). A point process driven multiple change-point model: A robust resistant approach. *Mathematical Biosciences.**220***(1), 57-71. doi:10.1016/j.mbs.2009.04.003

Bhattacharya, P. K. (1994). Some aspects of change-point analysis. Lecture Notes-Monograph Series, 28-56.

Bhattacharyya, G. K., & Johnson, R. A. (1968). Nonparametric Tests for Shift at an Unknown Time Point. *The Annals of Mathematical Statistics*, *39*(5), 1731–1743.

Chen, J., & Gupta, A. K. (2001). On change point detection and estimation. *Communications in* Statistics-Simulation and Computation, 30(3), 665–697.

Chen, J., & Gupta, A. K. (2011). Parametric statistical change point analysis: With applications to genetics, medicine, and finance. Springer Science & Business Media.

Csörgö, M., & Horváth, L. (1997). Limit theorems in change-point analysis (Vol. 18). John Wiley & Sons Inc.

Dabye, A. S., & Kutoyants, Y. A. (2001). Misspecified Change-Point Estimation Problem for a Poisson Process. *Journal of Applied Probability*, *38*, 122–130.

Fotopoulos et al. (2010). Exact asymptotic distribution of change-point mle for change in the mean of Gaussian sequences. *The Annals of Applied Statistics*, 4(2), 1081 - 1104. doi: 10.1214/09-AOAS294

Girshick, M. A., & Rubin, H. (1952). A Bayes approach to a quality control model. *The Annals of Mathematical Statistics*, 114–125.

Gurevich, G., & Vexler, A. (2010). Retrospective change point detection: from parametric to distribution free policies. *Communications in Statistics—Simulation and Computation®*, *39*(5), 899–920.

Hawkins, D. M., & Zamba, K. D. (2005). A change-point model for a shift in variance. *Journal of Quality Technology*, *37*(1), 21–31.

Hinkley, D. V. (1970). Inference About the Change-Point in a Sequence of Random Variables. *Biometrika*, *57*(1), 1–17.

Hinkley, D. V. (1971). Inference about the Change-Point from Cumulative Sum Tests. *Biometrika*, *58*(3), 509–523.

Jann, A. (2000). Multiple change-point detection with a genetic algorithm. *Soft Computing*, *4*(2), 68–75.

Karl, T. R., & Williams Jr, C. N. (1987). An approach to adjusting climatological time series for discontinuous inhomogeneities. *Journal of Climate and Applied Meteorology*, *26*(12), 1744–1763.

Khoo, M. B. (2004). Determining the time of a permanent shift in the process mean of CUSUM control charts. *Quality Engineering*, *17*(1), 87–93.

Page, E. S. (1954). Continuous Inspection Schemes. *Biometrika*, *41*(1/2), 100–115.

Page, E. S. (1955). A test for a change in a parameter occurring at an unknown point. *Biometrika*, *42*(3-4), 523–527. doi:10.1093/biomet/42.3-4.523

Page, E. S. (1957). On problems in which a change in a parameter occurs at an unknown point. *Biometrika*, *44*(1-2), 248–252. doi:10.1093/biomet/44.1-2.248

 ACCEPTED MANUSCRIPT

Reynolds, M.R., & Jianying, L. (2010). An evaluation of a GLR control chart for monitoring the process mean. *Journal of Quality Technology*, *42*(3), 287–310.

Samuel, T. R., & Pignatiello, J. J. (1998). Identifying the Time of a Change in a Poisson Rate Parameter. *Quality Engineering*, *10*(4), 673. doi:10.1080/08982119808919185

Samuel et al. (1998a). Identifying the Time of a Step Change in a Normal Process Variance. *Quality Engineering*, *10*(3), 529. doi:10.1080/08982119808919167

Samuel et al. (1998b). Identifying the Time of a Step Change with Control Charts. *Quality Engineering*, *10*(3), 521. doi:10.1080/08982119808919166

Shewhart, W. A. (1931). *Economic control of quality of manufactured product*. D. Van Nostrand Company, Inc.

Tercero, V. G. (2011). Nonparametric change-point estimation for observations following a random walk with drift. Retrieved from http://thinktech.lib.ttu.edu/ttu-ir/handle/2346/ETD-TTU-2011-05-1479

Tercero-Gómez, V. G., Garza-Venegas, J., & Cordero-Franco, A. (2013). Likelihood ratio test for a change-point in the mean of independent normal observations. IIE Annual Conference.Proceedings, , 3152-3159. Retrieved from http://search.proquest.com/docview/1471961772?accountid=11643

Wolfe, D. A., & Schechtman, E. (1984). Nonparametric statistical procedures for the changepoint problem. Journal of Statistical Planning and Inference, 9(3), 389-396.

ACCEPTED MANUSCRIPT

Table 1. For $\mu_0 \neq \mu_1$ and $\sigma_0 = \sigma_1$, given τ/T = 0.5, change-point estimations using $\hat{\tau}_{MLE1}$ and $\hat{\tau}_{CUSUM}$ are evaluated over different series length (T), sample sizes (n), and shifts of the mean (δ) measured in standard deviations. Estimations are presented next to their corresponding standard error, which are within parentheses.

T	n	$\hat{\tau}_{MLE1}$			$\hat{\tau}_{CUSUM}$		
		$\delta = 1$	$\delta = 1.5$	$\delta = 3$	$\delta = 1$	$\delta = 1.5$	$\delta = 3$
20	1	0.03 (2.49)	0.02 (1.81)	0.00 (0.59)	0.02 (2.49)	0.01 (1.50)	0.00 (0.45)
	3	0.03 (1.53)	0.00 (0.82)	0.00 (0.11)	0.02 (1.20)	0.00 (0.60)	0.00 (0.10)
	5	0.00 (1.06)	0.00 (0.44)	0.00 (0.03)	-0.01 (0.79)	0.00 (0.35)	0.00 (0.02)
30	1	-0.05 (3.96)	-0.01 (2.43)	-0.01 (0.58)	-0.01 (2.97)	-0.01 (1.69)	-0.01 (0.46)
	3	0.00 (1.95)	-0.01 (0.82)	0.00 (0.11)	0.00 (1.33)	0.00 (0.63)	0.00 (0.10)
	5	0.00 (1.13)	0.00 (0.43)	0.00 (0.03)	0.01 (0.83)	0.00 (0.36)	0.00 (0.02)
50	1	-0.04 (5.69)	0.03 (2.84)	0.00 (0.53)	-0.04 (3.61)	0.03 (1.93)	-0.01 (0.46)
	3	-0.02 (2.02)	0.00 (0.77)	0.00 (0.11)	-0.03 (1.49)	0.00 (0.66)	0.00 (0.10)
	5	-0.02 (1.05)	0.00 (0.43)	0.00 (0.03)	-0.02 (0.85)	0.00 (0.36)	0.00 (0.03)
100	1	0.01 (6.44)	-0.03 (2.64)	0.00 (0.53)	0.02 (4.12)	-0.01 (2.06)	0.00 (0.50)
	3	0.01 (1.82)	0.02 (0.75)	0.00 (0.10)	0.00 (1.51)	0.01 (0.69)	0.00 (0.10)
	5	0.00 (1.02)	-0.01 (0.42)	0.00 (0.03)	0.01 (0.94)	-0.01 (0.38)	0.00 (0.03)
300	1	0.03 (5.56)	-0.01 (2.30)	0.00 (0.49)	0.03 (4.57)	-0.01 (2.17)	0.00 (0.48)
	3	0.00 (1.73)	0.00 (0.71)	0.00 (0.09)	0.00 (1.62)	0.01 (0.69)	0.00 (0.09)
	5	-0.01 (0.98)	0.00 (0.38)	0.00 (0.03)	0.00 (0.94)	0.00 (0.37)	0.00 (0.03)
1000	1	0.09 (5.20)	0.01 (2.28)	-0.01 (0.50)	0.07 (4.93)	0.00 (2.23)	0.00 (0.50)
	3	-0.01 (1.63)	0.00 (0.70)	0.00 (0.10)	-0.01 (1.61)	0.01 (0.69)	0.00 (0.10)
	5	0.00 (1.00)	0.00 (0.38)	0.00 (0.03)	0.00 (0.98)	0.00 (0.38)	0.00 (0.03)

Table 2. For $\mu_0 \neq \mu_1$ and $\sigma_0 = \sigma_1$, given a fixed shift in the mean of δ = 1.5 standard deviations, performance of $\hat{\tau}_{MLE1}$ and $\hat{\tau}_{CUSUM}$ are evaluated over different series length (T), sample sizes (*n*), and change-point locations (τ / T). Estimations are presented next to their corresponding standard error, which are within parentheses.

T	*n*	$\hat{\tau}_{MLE1}$			$\hat{\tau}_{CUSUM}$		
		τ / T = 0.2	τ / T = 0.3	τ / T = 0.5	τ / T = 0.2	τ / T = 0.3	τ / T = 0.5
20	1	2.97 (2.91)	0.84 (2.10)	0.02 (1.81)	2.41 (3.18)	1.17 (2.20)	0.01 (1.50)
	3	1.12 (1.27)	0.14 (0.82)	0.00 (0.82)	1.15 (1.71)	0.45 (0.99)	0.00 (0.60)
	5	1.23 (0.78)	0.04 (0.43)	0.00 (0.44)	0.68 (1.27)	0.26 (0.66)	0.00 (0.35)
30	1	1.38 (3.64)	0.60 (2.78)	-0.01 (0.58)	2.77 (3.95)	1.35 (2.60)	-0.01 (0.46)
	3	0.15 (1.09)	0.03 (0.80)	0.00 (0.11)	1.31 (1.98)	0.51 (1.14)	0.00 (0.10)
	5	0.04 (0.45)	0.01 (0.45)	0.00 (0.03)	0.75 (1.44)	0.27 (0.70)	0.00 (0.02)
50	1	0.73 (3.82)	0.26 (3.04)	0.03 (2.84)	3.26 (4.86)	1.47 (2.99)	0.03 (1.93)
	3	0.38 (1.07)	0.01 (0.81)	0.00 (0.77)	1.81 (2.24)	0.53 (1.21)	0.00 (0.66)
	5	0.02 (0.44)	0.00 (0.42)	0.00 (0.43)	0.82 (1.59)	0.31 (0.80)	0.00 (0.36)
100	1	0.24 (3.22)	0.12 (2.70)	-0.03 (2.64)	3.96 (6.34)	1.72 (3.55)	-0.01 (2.06)
	3	0.33 (0.93)	0.02 (0.74)	0.02 (0.75)	1.96 (2.64)	0.57 (1.34)	0.01 (0.69)
	5	0.00 (0.42)	0.00 (0.38)	-0.01 (0.42)	0.83 (1.69)	0.30 (0.79)	-0.01 (0.38)
300	1	0.05 (2.40)	0.01 (2.37)	-0.01 (2.30)	4.47 (7.81)	1.86 (4.09)	-0.01 (2.17)
	3	0.31 (0.86)	0.00 (0.72)	0.00 (0.71)	2.14 (3.09)	0.60 (1.43)	0.01 (0.69)
	5	0.01 (0.39)	0.00 (0.38)	0.00 (0.38)	0.87 (1.84)	0.32 (0.83)	0.00 (0.37)
1000	1	0.03 (2.27)	-0.01 (2.33)	0.01 (2.28)	5.06 (9.33)	1.92 (4.26)	0.00 (2.23)
	3	0.30 (0.90)	-0.01 (0.70)	0.00 (0.70)	2.24 (3.33)	0.60 (1.42)	0.01 (0.69)
	5	0.00 (0.39)	0.00 (0.38)	0.00 (0.38)	0.90 (1.88)	0.32 (0.86)	0.00 (0.38)

Table 3. For $\mu_0 = \mu_1$ and $\sigma_0 \neq \sigma_1$, given a $\tau / \mathrm{T} = 0.5$, change-point estimations using $\hat{\tau}_{MLE2}$ and $\hat{\tau}_G$ are evaluated over different series length (T), sample sizes (n), and shifts in the ratio of deviations (σ_1/σ_0). Estimations are presented next to their corresponding standard error, which are within parentheses.

T	n	$\hat{\tau}_{MLE2}$			$\hat{\tau}_G$		
		$\sigma_1/\sigma_0 = 1.3$	$\sigma_1/\sigma_0 = 1.5$	$\sigma_1/\sigma_0 = 3$	$\sigma_1/\sigma_0 = 1.3$	$\sigma_1/\sigma_0 = 1.5$	$\sigma_1/\sigma_0 = 3$
20	1	0.35 (3.35)	0.41 (3.18)	0.41 (2.03)	0.28 (5.27)	0.31 (4.89)	0.43 (2.51)
	3	0.31 (3.01)	0.36 (2.54)	0.14 (0.85)	0.27 (4.63)	0.36 (3.75)	0.15 (0.88)
	5	0.26 (2.72)	0.26 (2.13)	0.06 (0.47)	0.29 (4.09)	0.28 (2.97)	0.06 (0.46)
30	1	0.55 (6.42)	0.68 (5.84)	0.58 (2.70)	0.50 (8.43)	0.66 (7.55)	0.61 (2.80)
	3	0.53 (5.36)	0.45 (4.12)	0.14 (0.86)	0.53 (6.94)	0.41 (5.04)	0.14 (0.85)
	5	0.43 (4.73)	0.38 (3.18)	0.05 (0.45)	0.41 (5.93)	0.42 (3.66)	0.06 (0.44)
50	1	1.15 (12.20)	1.10 (10.15)	0.76 (3.00)	0.88 (14.36)	0.88 (11.88)	0.78 (2.92)
	3	0.60 (9.24)	0.63 (5.99)	0.16 (0.81)	0.65 (10.69)	0.65 (6.54)	0.16 (0.80)
	5	0.74 (7.18)	0.45 (3.93)	0.06 (0.43)	0.63 (8.07)	0.44 (4.08)	0.06 (0.43)
100	1	1.67 (24.25)	1.48 (17.40)	0.80 (2.73)	1.27 (26.82)	1.18 (18.45)	0.81 (2.71)
	3	0.87 (14.51)	0.91 (6.86)	0.14 (0.75)	1.05 (15.47)	0.93 (6.96)	0.15 (0.76)
	5	0.68 (9.52)	0.51 (3.87)	0.05 (0.41)	0.71 (9.82)	0.50 (3.86)	0.05 (0.41)
300	1	2.43 (48.80)	3.21 (21.57)	0.79 (2.39)	2.52 (50.44)	3.20 (21.24)	0.80 (2.40)
	3	1.81 (16.24)	0.85 (5.84)	0.15 (0.72)	1.81 (16.04)	0.86 (5.84)	0.15 (0.72)
	5	0.95 (8.79)	0.49 (3.46)	0.05 (0.42)	0.94 (8.84)	0.49 (3.45)	0.05 (0.42)
1000	1	4.90 (47.74)	3.26 (17.08)	0.78 (2.35)	4.96 (47.25)	3.26 (17.23)	0.78 (2.35)
	3	1.58 (13.71)	0.92 (5.39)	0.16 (0.74)	1.57 (13.73)	0.92 (5.39)	0.16 (0.74)
	5	0.88 (7.81)	0.47 (3.14)	0.06 (0.42)	0.88 (7.80)	0.47 (3.13)	0.06 (0.42)

 ACCEPTED MANUSCRIPT

Table 4. For $\mu_0 = \mu_1$ and $\sigma_0 \neq \sigma_1$, given a fixed $\sigma_1/\sigma_0 = 1.5$, performance of $\hat{\tau}_{MLE2}$ and $\hat{\tau}_G$ are evaluated over different series length (T), sample sizes (n), and change-point locations (τ / T). Estimations are presented next to their corresponding standard error, which are within parentheses.

T	n	$\hat{\tau}_{MLE2}$			$\hat{\tau}_G$		
		τ / T = 0.2	τ / T = 0.3	τ / T = 0.5	τ / T = 0.2	τ / T = 0.3	τ / T = 0.5
20	1	5.57 (3.55)	3.39 (3.37)	0.41 (3.18)	5.37 (5.37)	3.51 (5.08)	0.31 (4.89)
	3	3.99 (3.43)	2.11 (2.96)	0.36 (2.54)	3.44 (4.83)	2.11 (4.18)	0.36 (3.75)
	5	3.88 (3.29)	1.41 (2.52)	0.26 (2.13)	2.47 (4.25)	1.35 (3.44)	0.28 (2.97)
30	1	6.83 (6.71)	4.57 (6.20)	0.68 (5.84)	7.21 (8.46)	4.74 (7.95)	0.66 (7.55)
	3	3.71 (5.64)	2.25 (4.75)	0.45 (4.12)	3.87 (6.91)	2.40 (5.80)	0.41 (5.04)
	5	2.41 (4.61)	1.25 (3.57)	0.38 (3.18)	2.40 (5.56)	1.29 (4.29)	0.42 (3.66)
50	1	9.74 (12.66)	6.15 (11.11)	1.10 (10.15)	10.24 (14.51)	6.36 (12.87)	0.88 (11.88)
	3	4.15 (8.54)	2.10 (6.78)	0.63 (5.99)	4.54 (9.88)	2.23 (7.61)	0.65 (6.54)
	5	1.83 (5.76)	0.97 (4.37)	0.45 (3.93)	1.87 (6.40)	1.06 (4.78)	0.44 (4.08)
100	1	11.88 (23.32)	7.08 (19.50)	1.48 (17.40)	12.78 (25.36)	7.44 (21.20)	1.18 (18.45)
	3	3.08 (10.48)	1.57 (7.97)	0.91 (6.86)	3.27 (11.16)	1.64 (8.14)	0.93 (6.96)
	5	1.06 (5.35)	0.67 (4.11)	0.51 (3.87)	1.11 (5.54)	0.67 (4.09)	0.50 (3.86)
300	1	8.77 (34.84)	4.90 (24.65)	3.21 (21.57)	9.19 (35.57)	5.27 (25.45)	3.20 (21.24)
	3	1.57 (6.89)	1.05 (6.04)	0.85 (5.84)	1.63 (6.90)	1.05 (6.00)	0.86 (5.84)
	5	0.53 (3.40)	0.54 (3.56)	0.49 (3.46)	0.54 (3.43)	0.55 (3.57)	0.49 (3.45)
1000	1	3.83 (20.22)	3.38 (18.02)	3.26 (17.08)	4.04 (20.35)	3.48 (17.96)	3.26 (17.23)
	3	1.40 (5.57)	0.94 (5.40)	0.92 (5.39)	1.41 (5.61)	0.96 (5.44)	0.92 (5.39)
	5	0.45 (3.23)	0.46 (3.25)	0.47 (3.14)	0.47 (3.24)	0.46 (3.25)	0.47 (3.13)

 ACCEPTED MANUSCRIPT

Table 5. For $\mu_0 \neq \mu_1$ and $\sigma_0 \neq \sigma_1$, given T = 30 and τ / T = 0.5, change-point estimations using $\hat{\tau}_{MLE3}$ are evaluated in part (a), $\hat{\tau}_{CUSUM}$ in part (b), and $\hat{\tau}_{G}$ in part (c) over different ratios (σ_1/σ_0), sample sizes (n), and shifts in the mean (δ). Estimations are presented next to their corresponding standard error, which are within parentheses.

Part (a). Performance of the MLE $\hat{\tau}_{MLE3}$					
σ_1/σ_0	n	$\delta = 0$	$\delta = 1$	$\delta = 1.5$	$\delta = 3$
1.0	1	See Table 7	-0.02 (4.60)	0.05 (2.97)	0.00 (0.66)
	3		-0.02 (2.15)	0.00 (0.93)	0.00 (0.11)
	5		0.02 (1.23)	0.00 (0.44)	0.00 (0.04)
1.5	1	0.81 (6.12)	0.40 (4.72)	0.35 (3.49)	0.10 (1.10)
	3	0.50 (4.49)	0.24 (2.21)	0.12 (1.21)	0.01 (0.22)
	5	0.34 (3.40)	0.14 (1.30)	0.05 (0.59)	0.00 (0.09)
3.0	1	0.57 (3.02)	0.54 (2.81)	0.55 (2.48)	0.38 (1.57)
	3	0.15 (0.88)	0.13 (0.76)	0.12 (0.68)	0.05 (0.37)
	5	0.05 (0.47)	0.05 (0.38)	0.03 (0.32)	0.01 (0.16)

Part (b). Performance of the CUSUM estimator $\hat{\tau}_{CUSUM}$.					
σ_1/σ_0	n	$\delta = 0$	$\delta = 1$	$\delta = 1.5$	$\delta = 3$
1.0	1	See Table 7	0.01 (2.98)	0.01 (1.71)	0.01 (0.46)
	3		-0.01 (1.33)	0.00 (0.63)	0.00 (0.09)
	5		0.01 (0.85)	0.01 (0.35)	0.00 (0.02)
1.5	1	2.79	1.23	0.72	0.20 (0.78)

 ACCEPTED MANUSCRIPT

		(5.98)	(3.75)	(2.38)	
	3	2.93 (5.98)	0.54 (1.91)	0.26 (1.01)	0.04 (0.23)
	5	2.86 (5.99)	0.37 (1.33)	0.16 (0.63)	0.01 (0.11)
3.0	1	5.40 (4.35)	3.94 (4.19)	2.88 (3.72)	1.08 (1.95)
	3	5.28 (4.38)	2.49 (3.43)	1.39 (2.34)	0.38 (0.86)
	5	5.24 (4.40)	1.74 (2.78)	0.87 (1.64)	0.21 (0.57)
Part (c). Performance of the Hawkins' estimator $\hat{\tau}_G$.					
σ_1/σ_0	n	$\delta = 0$	$\delta = 1$	$\delta = 1.5$	$\delta = 3$
1.0	1	See Table 7	0.00 (8.95)	0.10 (8.63)	0.11 (7.24)
	3		-0.05 (8.65)	0.04 (8.21)	0.00 (6.62)
	5		-0.04 (8.59)	0.02 (8.05)	0.05 (6.36)
1.5	1	0.65 (7.50)	-0.31 (7.46)	-1.34 (7.18)	-2.75 (6.05)
	3	0.52 (5.02)	-1.28 (4.85)	-2.55 (4.51)	-4.27 (3.24)
	5	0.32 (3.63)	-1.30 (3.46)	-2.71 (3.20)	-4.46 (2.14)
3.0	1	0.59 (2.83)	0.33 (2.79)	0.14 (2.81)	-0.92 (2.64)
	3	0.16 (0.84)	0.07 (0.79)	-0.01 (0.80)	-0.48 (1.05)
	5	0.06 (0.46)	0.02 (0.40)	-0.03 (0.41)	-0.31 (0.74)

Table 6. For $\mu_0 \neq \mu_1$ and $\sigma_0 \neq \sigma_1$, given a fixed $\sigma_1/\sigma_0 = 1.5$, and a shift in mean of $\delta = 1.5$, performance of $\hat{\tau}_{MLE3}$ is evaluated in part (a) and $\hat{\tau}_{CUSUM}$ in part (b), and $\hat{\tau}_G$ in part (c) over different series length (T), sample sizes (n), and change-point locations (τ / T). Estimations are presented next to their corresponding standard error, which are within parentheses.

Part (a). Performance of the MLE $\hat{\tau}_{MLE3}$.				
T	n	τ / T = 0.2	τ / T = 0.3	τ / T = 0.5
20	1	4.26 (3.49)	1.64 (2.74)	0.24 (2.28)
	3	1.81 (2.22)	0.34 (1.24)	0.12 (1.09)
	5	1.79 (1.80)	0.12 (0.65)	0.06 (0.65)
30	1	3.03 (5.34)	1.58 (4.11)	0.38 (3.52)
	3	0.38 (1.85)	0.23 (1.37)	0.12 (1.19)
	5	0.13 (0.80)	0.08 (0.68)	0.05 (0.60)
50	1	2.38 (7.06)	1.23 (5.22)	0.57 (4.16)
	3	0.47 (1.57)	0.18 (1.24)	0.14 (1.10)
	5	0.07 (0.63)	0.06 (0.61)	0.05 (0.58)
100	1	1.19 (6.53)	0.79 (4.23)	0.62 (3.74)
	3	0.38 (1.19)	0.15 (1.09)	0.12 (0.97)
	5	0.06 (0.56)	0.06 (0.58)	0.05 (0.55)
1000	1	0.65 (3.02)	0.65 (3.01)	0.58 (3.00)
	3	0.35 (1.07)	0.13 (0.94)	0.12 (0.93)
	5	0.05 (0.52)	0.04 (0.53)	0.05 (0.52)
Part (b). Performance of the CUSUM estimator $\hat{\tau}_{CUSUM}$.				
T	n	τ / T = 0.2	τ / T = 0.3	τ / T = 0.5
20	1	3.74 (3.81)	2.21 (2.96)	0.58 (2.07)
	3	1.89 (2.47)	0.99 (1.65)	0.24 (0.91)
	5	1.32 (2.00)	0.64 (1.19)	0.15 (0.60)
30	1	4.57 (5.11)	2.57 (3.68)	0.71 (2.41)
	3	2.23 (3.05)	1.13 (1.91)	0.26 (0.99)
	5	1.52 (2.40)	0.72 (1.37)	0.15 (0.62)
50	1	5.61 (6.97)	3.03 (4.64)	0.79 (2.69)
	3	2.99 (3.81)	1.24 (2.22)	0.29 (1.08)
	5	1.69 (2.77)	0.76 (1.50)	0.16 (0.66)
100	1	7.17 (9.92)	3.73 (5.98)	0.91 (3.17)

 ACCEPTED MANUSCRIPT

	3	3.54 (4.81)	1.43 (2.62)	0.31 (1.17)
	5	1.97 (3.42)	0.86 (1.73)	0.18 (0.70)
1000	1	11.27 (19.42)	4.79 (8.68)	1.15 (3.81)
	3	4.30 (7.01)	1.60 (3.17)	0.35 (1.24)
	5	2.25 (4.27)	0.89 (1.86)	0.19 (0.73)
Part (c). Performance of the Hawkins' estimator $\hat{\tau}_G$.				
T	n	τ / T = 0.2	τ / T = 0.3	τ / T = 0.5
20	1	5.00 (5.50)	3.03 (5.22)	-0.52 (4.73)
	3	2.60 (5.01)	1.04 (4.24)	-1.44 (3.46)
	5	1.63 (4.27)	0.08 (3.21)	-1.74 (2.61)
30	1	6.69 (8.80)	3.76 (8.15)	-1.32 (7.18)
	3	2.49 (6.99)	0.43 (5.64)	-2.56 (4.48)
	5	1.19 (5.52)	-0.61 (3.75)	-2.71 (3.18)
50	1	9.04 (15.02)	3.95 (13.26)	-3.09 (11.11)
	3	2.25 (10.00)	-1.04 (6.84)	-4.49 (5.56)
	5	-0.01 (6.01)	-1.62 (3.82)	-4.09 (4.06)
100	1	9.88 (27.16)	0.78 (20.87)	-8.80 (16.44)
	3	-0.52 (10.15)	-3.36 (6.70)	-8.00 (7.64)
	5	-1.37 (4.27)	-2.75 (4.14)	-6.88 (5.98)
1000	1	-9.82 (19.07)	-19.08 (26.60)	-61.68 (46.20)
	3	-4.16 (7.97)	-10.69 (14.85)	-55.11 (32.71)
	5	-2.90 (5.42)	-7.87 (11.05)	-53.52 (27.53)

 ACCEPTED MANUSCRIPT

Table 7. For $\mu_0 = \mu_1$ and $\sigma_0 =| \sigma_1$ (that is, no change-point has occurred) performance of $\hat{\tau}_{MLE1}$, $\hat{\tau}_{MLE2}$, $\hat{\tau}_{MLE3}$, $\hat{\tau}_{CUSUM}$ and $\hat{\tau}_G$ were evaluated over different series length (T) and sample sizes (n). Estimations are presented next to their corresponding standard deviation, which are within parentheses.

T	n	Estimator				
		$\hat{\tau}_{MLE1}$	$\hat{\tau}_{MLE2}$	$\hat{\tau}_{MLE3}$	$\hat{\tau}_{CUSUM}$	$\hat{\tau}_G$
20	1	9.98 (3.48)	9.93 (3.53)	9.99 (3.58)	10.01 (4.37)	9.99 (5.57)
	3	10.01 (3.47)	10.00 (3.51)	9.97 (3.53)	9.96 (4.36)	9.99 (5.55)
	5	9.98 (3.47)	10.09 (3.48)	10.02 (3.51)	9.97 (4.38)	10.03 (5.53)
30	1	14.87 (6.81)	14.87 (6.88)	14.84 (7.03)	14.93 (6.54)	14.97 (9.11)
	3	14.98 (6.83)	15.04 (6.82)	14.95 (6.90)	15.12 (6.57)	15.04 (9.04)
	5	15.05 (6.77)	15.02 (6.83)	15.03 (6.86)	15.02 (6.52)	14.84 (9.12)
50	1	24.96 (13.78)	24.93 (13.92)	25.03 (14.37)	24.94 (10.92)	25.08 (16.57)
	3	24.90 (13.76)	25.01 (13.73)	24.79 (13.92)	24.90 (10.94)	25.10 (16.26)
	5	24.93 (13.75)	24.97 (13.73)	25.03 (13.89)	24.90 (10.95)	24.98 (16.28)
100	1	49.65 (31.73)	50.51 (32.24)	50.01 (33.26)	49.92 (21.73)	50.25 (35.72)
	3	50.44 (31.89)	50.18 (31.84)	50.26 (32.36)	50.05 (21.76)	49.96 (35.20)
	5	49.43 (31.78)	49.51 (31.83)	49.70 (32.17)	49.67 (21.88)	49.86 (35.10)

Table 8. For $\mu_0 \neq \mu_1$ and $\sigma_0 = \sigma_1$, change-point estimations using $\hat{\tau}_{MLE1}$ and $\hat{\tau}_{CUSUM}$ for t distribution with 3 degrees of freedom (t_3) and a gamma with α=0.5 and β=1 ($G(0.5,1)$) are evaluated over different series length (T), sample sizes (n), change-point locations (τ/T) and shifts of the mean (δ) measured in standard deviations. Estimations are presented next to their corresponding standard error, which are within parentheses.

t_3		$\hat{\tau}_{MLE1}$				$\hat{\tau}_{CUSUM}$			
		τ/T=0.2		τ/T=0.5		τ/T=0.2		τ/T=0.5	
T	n	$\delta = 1$	$\delta = 1.5$	$\delta = 1$	$\delta = 1.5$	$\delta = 1$	$\delta = 1.5$	$\delta = 1$	$\delta = 1.5$
20	1	3.75 (3.39)	2.59 (2.75)	0.00 (2.23)	-0.02 (1.53)	3.25 (3.91)	2.00 (3.02)	0.00 (2.32)	0.00 (1.38)
	3	1.74 (2.22)	1.03 (1.20)	0.03 (1.41)	0.00 (0.76)	1.72 (2.58)	1.03 (1.61)	0.01 (1.22)	0.00 (0.67)
	5	1.72 (1.73)	1.20 (0.79)	0.01 (0.98)	0.00 (0.45)	1.18 (2.03)	0.55 (1.24)	0.01 (0.78)	0.00 (0.40)
30	1	3.05 (5.51)	1.10 (3.36)	0.02 (3.62)	0.01 (2.28)	4.13 (5.38)	2.26 (3.76)	0.04 (2.86)	0.02 (1.78)
	3	0.75 (2.80)	0.15 (1.32)	-0.02 (1.81)	0.01 (0.85)	2.08 (3.30)	1.16 (1.92)	-0.03 (1.26)	0.00 (0.72)
	5	0.32 (1.57)	0.04 (0.49)	0.01 (1.17)	0.00 (0.50)	1.35 (2.42)	0.64 (1.41)	0.01 (0.83)	0.00 (0.42)
50	1	2.78 (8.00)	0.67 (3.98)	-0.07 (5.23)	0.05 (2.74)	5.01 (7.34)	2.65 (4.60)	-0.03 (3.38)	0.00 (1.90)
	3	0.39 (2.88)	0.03 (1.36)	-0.02 (2.04)	0.00 (0.94)	2.43 (3.85)	1.32 (2.23)	0.01 (1.48)	-0.01 (0.72)
	5	0.13 (1.47)	0.02 (0.67)	-0.01 (1.25)	0.00 (0.54)	1.53 (2.84)	0.67 (1.54)	-0.01 (0.94)	0.00 (0.46)
100	1	1.82 (10.12)	0.42 (4.50)	-0.08 (6.88)	-0.04 (3.25)	6.44 (10.30)	3.50 (6.55)	-0.07 (4.17)	0.00 (2.30)
	3	0.14 (3.04)	0.00 (1.43)	0.02 (2.18)	0.00 (1.18)	2.90 (4.84)	1.52 (2.68)	0.02 (1.68)	0.01 (0.80)
	5	0.03 (1.21)	0.01 (0.52)	-0.02 (1.39)	0.00 (0.49)	1.71 (3.33)	0.72 (1.66)	0.00 (0.97)	0.00 (0.47)

$G(0.5,1)$	$\hat{\tau}_{MLE1}$		$\hat{\tau}_{CUSUM}$	
	τ/T=0.2	τ/T=0.5	τ/T=0.2	τ/T=0.5

T	n	$\delta = 1$	$\delta = 1.5$	$\delta = 1$	$\delta = 1.5$	$\delta = 1$	$\delta = 1.5$	$\delta = 1$	$\delta = 1.5$
20	1	3.91 (3.39)	2.78 (2.99)	-0.15 (2.40)	-0.26 (1.65)	3.01 (3.82)	2.09 (3.13)	-0.53 (2.49)	-0.42 (1.49)
	3	1.90 (2.37)	0.95 (1.16)	-0.18 (1.47)	-0.20 (0.75)	1.80 (2.59)	1.07 (1.60)	-0.22 (1.20)	-0.16 (0.58)
	5	1.82 (1.91)	1.12 (0.66)	-0.16 (1.02)	-0.11 (0.43)	1.30 (2.11)	0.60 (1.19)	-0.15 (0.78)	-0.09 (0.35)
30	1	3.16 (5.68)	1.14 (3.86)	-0.32 (3.89)	-0.46 (2.33)	4.00 (5.27)	2.47 (3.91)	-0.63 (3.00)	-0.46 (1.55)
	3	0.73 (2.71)	0.07 (1.13)	-0.26 (1.95)	-0.21 (0.76)	2.07 (3.14)	1.20 (1.86)	-0.22 (1.30)	-0.18 (0.58)
	5	0.19 (1.47)	-0.06 (0.28)	-0.19 (1.14)	-0.10 (0.41)	1.48 (2.49)	0.66 (1.30)	-0.16 (0.82)	-0.08 (0.34)
50	1	2.74 (8.39)	0.33 (4.35)	-0.57 (5.78)	-0.65 (2.78)	5.29 (7.44)	3.06 (4.91)	-0.67 (3.64)	-0.57 (1.74)
	3	0.13 (2.78)	-0.10 (1.06)	-0.33 (1.99)	-0.23 (0.74)	2.57 (4.03)	1.38 (2.17)	-0.27 (1.36)	-0.20 (0.62)
	5	-0.10 (1.15)	-0.10 (0.39)	-0.21 (1.10)	-0.10 (0.41)	1.64 (2.82)	0.67 (1.39)	-0.17 (0.84)	-0.09 (0.38)
100	1	1.49 (10.91)	-0.47 (3.71)	-0.77 (7.15)	-0.77 (2.60)	7.16 (10.72)	3.65 (6.15)	-0.69 (4.12)	-0.63 (1.81)
	3	-0.15 (2.35)	-0.14 (0.91)	-0.34 (1.77)	-0.23 (0.76)	3.05 (4.87)	1.45 (2.36)	-0.29 (1.48)	-0.21 (0.66)
	5	-0.19 (1.04)	-0.10 (0.40)	-0.22 (0.99)	-0.11 (0.42)	1.88 (3.33)	0.69 (1.40)	-0.21 (0.92)	-0.10 (0.39)

Table 9. For $\mu_0 = \mu_1$ and $\sigma_0 \neq \sigma_1$, change-point estimations using $\hat{\tau}_{MLE2}$ and $\hat{\tau}_G$ for t distribution with 3 degrees of freedom (t_3) and a gamma with α =0.5 and β =1 ($G(0.5,1)$), are evaluated over different series length (T), sample sizes (n), change-point locations (τ/T) and ratios (σ_1/σ_0). Estimations are presented next to their corresponding standard error, which are within parentheses.

t_3		$\hat{\tau}_{MLE2}$				$\hat{\tau}_G$			
		τ/T =0.2		τ/T =0.5		τ/T =0.2		τ/T =0.5	
T	n	σ_1/σ_0 =1.3	σ_1/σ_0 =1.5	σ_1/σ_0 =1.3	σ_1/σ_0 =1.5	σ_1/σ_0 =1.3	σ_1/σ_0 =1.5	σ_1/σ_0 =1.3	σ_1/σ_0 =1.5
20	1	5.90 (3.57)	5.71 (3.56)	0.20 (3.52)	0.27 (3.39)	5.83 (5.02)	5.50 (4.97)	0.23 (4.95)	0.21 (4.76)
	3	5.37(3.56)	4.99 (3.54)	0.25 (3.37)	0.29 (3.20)	5.07 (5.08)	4.51 (4.91)	0.06 (4.91)	0.11 (4.56)
	5	5.59(3.55)	5.13 (3.57)	0.18 (3.32)	0.25 (3.03)	5.03 (5.15)	4.36 (4.99)	-0.01 (4.83)	0.00 (4.30)
30	1	8.52 (6.81)	7.85 (6.80)	0.21 (6.72)	0.30 (6.39)	8.59 (8.00)	7.99 (7.82)	0.17 (7.88)	0.28 (7.43)
	3	7.64 (6.74)	6.57 (6.64)	0.18 (6.34)	0.16 (5.77)	7.49 (8.04)	6.38 (7.74)	0.00 (7.63)	-0.06 (6.90)
	5	7.57 (6.79)	6.16 (6.60)	0.08 (6.12)	0.12 (5.50)	7.30 (8.10)	5.94 (7.66)	-0.14 (7.39)	-0.13 (6.56)
50	1	13.62 (12.99)	12.17 (12.92)	0.10 (12.80)	0.28 (12.06)	13.73 (13.82)	12.38 (13.49)	0.19 (13.57)	0.37 (12.64)
	3	11.88 (12.74)	9.49 (12.29)	0.03 (11.96)	0.24 (10.47)	11.73 (13.80)	9.25 (13.04)	-0.21 (13.15)	0.00 (11.24)
	5	11.34 (12.81)	8.77 (12.01)	-0.03 (11.48)	0.05 (9.78)	11.13 (13.88)	8.57 (12.78)	-0.27 (12.53)	-0.07 (10.55)
100	1	25.41 (28.64)	20.99 (27.26)	-0.21 (27.59)	0.40 (24.52)	25.37 (28.60)	21.31 (27.18)	-0.25 (27.69)	0.33 (24.34)
	3	21.01 (27.53)	15.49 (25.13)	-0.82 (25.23)	0.06 (20.56)	21.20 (28.23)	15.50 (25.53)	-0.86 (25.80)	-0.12 (20.88)
	5	19.20 (26.87)	12.83 (23.48)	-0.54 (23.60)	0.09 (18.65)	19.06 (27.71)	12.80 (23.90)	-0.62 (24.35)	0.01 (18.94)

$G(0.5,1)$	$\hat{\tau}_{MLE2}$		$\hat{\tau}_G$	
	τ/T =0.2	τ/T =0.5	τ/T =0.2	τ/T =0.5

T	n	σ_1/σ_0 =1.3	σ_1/σ_0 =1.5	σ_1/σ_0 =1.3	σ_1/σ_0 =1.5	σ_1/σ_0 =1.3	σ_1/σ_0 =1.5	σ_1/σ_0 =1.3	σ_1/σ_0 =1.5
20	1	5.93 (3.62)	5.92 (3.63)	0.08 (3.60)	0.01 (3.52)	6.04 (4.94)	5.83 (4.96)	0.10 (4.95)	0.09 (4.85)
	3	5.50 (3.52)	5.29 (3.54)	0.12 (3.46)	0.26 (3.34)	5.38 (5.22)	5.06 (5.13)	0.04 (5.16)	0.21 (4.92)
	5	5.79 (3.53)	5.55 (3.54)	0.24 (3.36)	0.30 (3.18)	5.52 (5.26)	4.96 (5.19)	0.21 (5.13)	0.22 (4.75)
30	1	8.52 (6.80)	8.33 (6.84)	-0.04 (6.69)	-0.04 (6.54)	8.85 (7.97)	8.69 (7.96)	0.25 (7.85)	0.29 (7.64)
	3	8.02 (6.64)	7.03 (6.56)	0.12 (6.48)	0.18 (6.14)	8.04 (8.31)	7.20 (8.09)	0.12 (8.16)	0.25 (7.61)
	5	7.91 (6.73)	6.91 (6.64)	0.26 (6.34)	0.41 (5.77)	7.95 (8.38)	6.85 (8.09)	0.05 (8.04)	0.28 (7.21)
50	1	13.59 (12.78)	12.56 (12.57)	-0.56 (12.64)	-0.73 (12.31)	14.56 (14.10)	13.73 (13.77)	0.26 (13.93)	0.38 (13.38)
	3	12.65 (12.83)	10.65 (12.38)	-0.07 (12.20)	0.30 (11.06)	12.77 (14.65)	10.99 (14.02)	0.31 (13.96)	0.28 (12.59)
	5	12.43 (13.00)	9.40 (12.04)	0.07 (11.91)	0.18 (10.19)	12.35 (14.76)	9.68 (13.62)	0.09 (13.60)	0.17 (11.48)
100	1	25.42 (27.19)	21.99 (26.58)	-1.32 (27.04)	-0.92 (25.36)	27.31 (29.56)	24.26 (28.72)	-0.14 (29.22)	0.36 (26.98)
	3	22.40 (27.78)	16.63 (25.11)	-0.36 (25.46)	0.48 (21.59)	23.00 (30.06)	17.31 (27.05)	0.02 (27.58)	0.68 (22.82)
	5	20.30 (27.63)	12.86 (23.15)	0.00 (24.02)	0.98 (18.12)	20.84 (29.85)	13.28 (24.83)	0.02 (25.90)	0.79 (19.07)

Table 10. For $\mu_0 \neq \mu_1$ and $\sigma_0 \neq \sigma_1$, given a T = 30 and a t distribution with 3 degrees of freedom (t_3) and a gamma with α =0.5 and β =1 ($G(0.5,1)$), change-point estimations using $\hat{\tau}_{MLE3}$ are evaluated in parts (a,d), $\hat{\tau}_{CUSUM}$ in parts (b,e), and $\hat{\tau}_G$ in parts (c,f) over different ratios (σ_1/σ_0), sample sizes (n), and shifts in the mean (δ). Mean values are next to their corresponding standard error within parentheses.

Part (a). Performance of the MLE $\hat{\tau}_{MLE3}$							
t_3		τ/T =0.2			τ/T =0.5		
σ_1/σ_0	n	$\delta = 0.0$	$\delta = 1$	$\delta = 1.5$	$\delta = 0.0$	$\delta = 1$	$\delta = 1.5$
1.0	1	15.04 (7.00)	4.38 (6.39)	2.22 (4.93)	15.02 (7.01)	0.04 (4.26)	0.04 (2.82)
	3	14.61 (6.86)	2.19 (4.89)	0.91 (3.21)	15.08 (6.84)	-0.03 (2.79)	-0.01 (1.41)
	5	14.94 (6.85)	1.71 (4.42)	0.63 (2.65)	14.91 (6.87)	0.00 (2.23)	-0.02 (1.11)
1.5	1	8.23 (6.83)	5.15 (6.59)	3.15 (5.64)	0.55 (6.40)	0.12 (4.68)	-0.03 (3.32)
	3	6.59 (6.65)	2.90 (5.44)	1.34 (4.16)	0.38 (5.82)	-0.18 (3.28)	-0.25 (1.98)
	5	6.32 (6.66)	2.23 (4.98)	0.90 (3.40)	0.23 (5.47)	-0.22 (2.64)	-0.20 (1.56)
3.0	1	4.55 (6.03)	4.01 (5.89)	3.27 (5.49)	0.68 (4.32)	0.49 (3.81)	0.22 (3.33)
	3	2.16 (4.60)	1.55 (4.07)	1.13 (3.83)	0.14 (2.83)	-0.02 (2.37)	-0.17 (1.99)
	5	1.57 (4.03)	1.08 (3.50)	0.79 (3.15)	-0.01 (2.25)	-0.16 (1.91)	-0.22 (1.55)
Part (b). Performance of the CUSUM estimator $\hat{\tau}_{CUSUM}$							
t_3		τ/T =0.2			τ/T =0.5		
σ_1/σ_0	n	$\delta = 0.0$	$\delta = 1$	$\delta = 1.5$	$\delta = 0.0$	$\delta = 1$	$\delta = 1.5$
1.0	1	15.01 (6.72)	3.96 (5.26)	2.37 (3.82)	15.02 (6.73)	0.05 (2.84)	0.00 (1.70)
	3	14.66 (6.68)	2.02 (3.13)	1.22 (1.99)	15.02 (6.62)	0.00 (1.33)	0.01 (0.67)

	5	14.87 (6.65)	1.40 (2.51)	0.64 (1.37)	15.05 (6.61)	-0.01 (0.88)	-0.01 (0.41)
1.5	1	10.23 (6.18)	6.14 (6.08)	4.02 (4.96)	2.54 (6.22)	0.99 (3.59)	0.53 (2.32)
	3	10.09 (6.12)	3.50 (4.45)	2.10 (3.06)	2.63 (6.17)	0.45 (1.89)	0.19 (1.01)
	5	10.45 (6.09)	2.50 (3.60)	1.41 (2.38)	2.65 (6.04)	0.29 (1.28)	0.13 (0.72)
3.0	1	11.26 (5.71)	9.58 (6.24)	7.83 (6.22)	5.11 (4.70)	3.52 (4.30)	2.44 (3.68)
	3	10.92 (5.64)	7.16 (6.04)	4.83 (5.21)	5.11 (4.55)	2.15 (3.30)	1.23 (2.32)
	5	11.31 (5.62)	5.91 (5.75)	3.79 (4.59)	5.13 (4.56)	1.65 (2.78)	0.81 (1.68)

Part (c). Performance of the Hawkins' estimator $\hat{\tau}_G$.

t_3		τ/T =0.2			τ/T =0.5		
σ_1/σ_0	n	$\delta = 0.0$	$\delta = 1$	$\delta = 1.5$	$\delta = 0.0$	$\delta = 1$	$\delta = 1.5$
1.0	1	14.99 (8.09)	8.79 (8.21)	8.47 (7.86)	15.06 (8.19)	0.00 (7.95)	-0.01 (7.52)
	3	14.62 (8.32)	8.77 (8.25)	8.49 (7.46)	14.94 (8.35)	0.00 (8.06)	0.05 (7.62)
	5	14.93 (8.41)	9.15 (8.28)	8.86 (7.26)	14.84 (8.41)	0.11 (8.14)	-0.04 (7.60)
1.5	1	8.13 (7.84)	7.68 (8.23)	7.45 (8.33)	0.21 (7.38)	-0.58 (7.23)	-1.28 (6.89)
	3	6.28 (7.72)	5.76 (8.09)	5.44 (8.24)	0.08 (6.85)	-1.37 (6.52)	-2.52 (5.93)
	5	5.86 (7.64)	5.19 (8.05)	4.71 (8.01)	-0.07 (6.53)	-1.79 (6.06)	-2.93 (5.46)
3.0	1	4.56 (6.40)	4.39 (6.52)	4.22 (6.69)	0.56 (4.49)	0.08 (4.42)	-0.45 (4.16)
	3	2.01 (4.89)	1.67 (4.74)	1.35 (4.76)	0.10 (3.04)	-0.23 (2.90)	-0.55 (2.71)
	5	1.46 (4.32)	1.12 (4.11)	0.88 (3.98)	-0.04 (2.48)	-0.30 (2.39)	-0.52 (2.20)

Part (d). Performance of the MLE $\hat{\tau}_{MLE3}$

$G(0.5,1)$		τ/T =0.2			τ/T =0.5		
σ_1/σ_0	n	$\delta = 0.0$	$\delta = 1$	$\delta = 1.5$	$\delta = 0.0$	$\delta = 1$	$\delta = 1.5$
1.0	1	14.91 (6.98)	5.84 (7.04)	3.37 (6.05)	14.88 (6.96)	-0.05 (5.02)	-0.40 (3.75)

 ACCEPTED MANUSCRIPT

	3	14.73 (6.95)	3.44 (5.80)	1.28 (3.95)	15.01 (6.85)	-0.16 (3.67)	-0.32 (2.06)
	5	15.03 (6.89)	2.28 (4.99)	0.52 (2.60)	14.94 (6.91)	-0.22 (2.70)	-0.19 (1.22)
1.5	1	8.64 (6.88)	7.07 (6.88)	5.33 (6.81)	0.48 (6.69)	0.60 (5.74)	-0.02 (4.67)
	3	7.58 (6.72)	4.99 (6.36)	2.73 (5.32)	0.62 (6.24)	0.20 (4.64)	-0.30 (3.07)
	5	7.15 (6.66)	3.97 (5.89)	1.65 (4.29)	0.55 (5.82)	0.04 (3.88)	-0.34 (2.19)
3.0	1	6.40 (6.92)	6.43 (6.43)	5.85 (6.27)	0.51 (5.05)	1.29 (5.26)	1.24 (4.97)
	3	3.37 (5.52)	3.23 (5.07)	2.75 (4.70)	0.17 (3.54)	0.82 (3.57)	0.63 (3.22)
	5	2.16 (4.50)	2.05 (3.99)	1.62 (3.57)	0.10 (2.62)	0.55 (2.65)	0.42 (2.32)

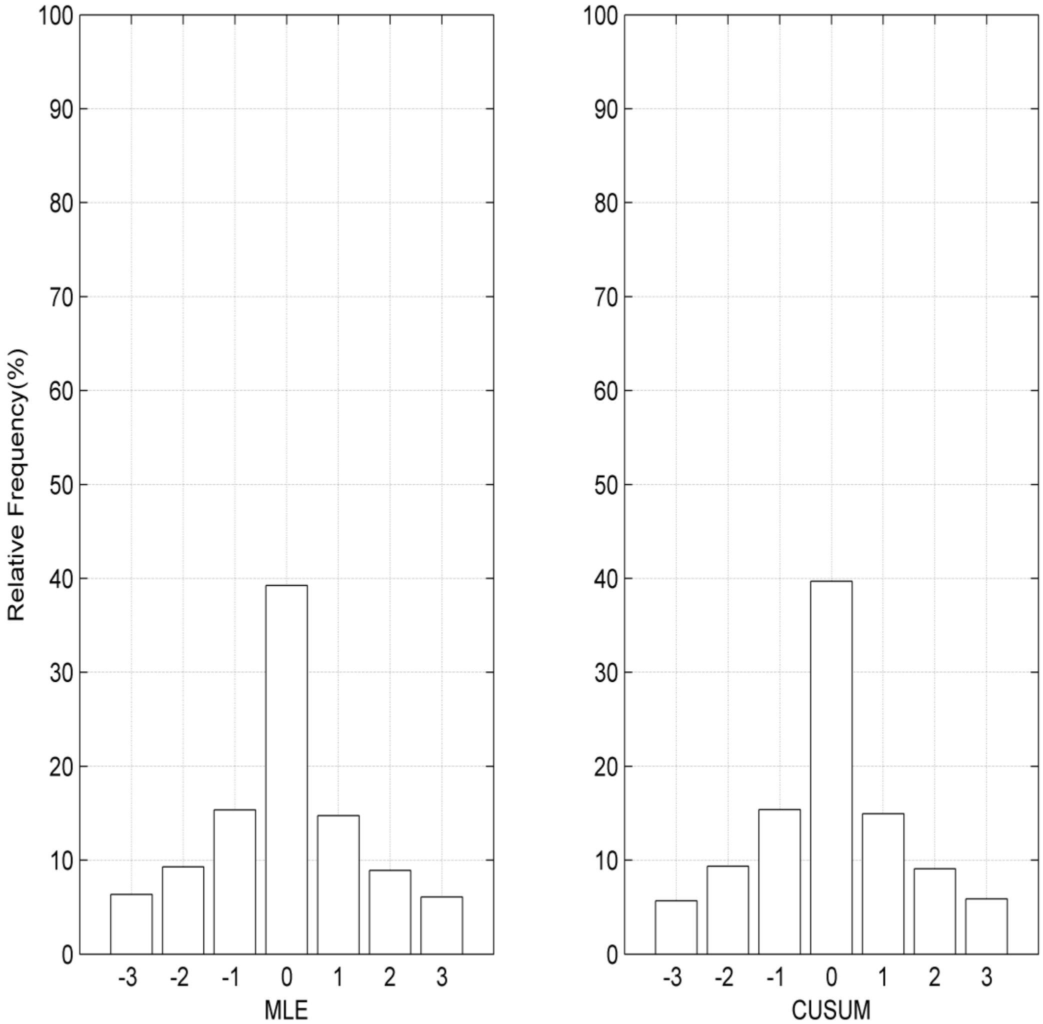
Part (e). Performance of the CUSUM estimator $\hat{\tau}_{CUSUM}$

$G(0.5,1)$		τ/T =0.2			τ/T =0.5		
σ_1/σ_0	n	$\delta = 0.0$	$\delta = 1$	$\delta = 1.5$	$\delta = 0.0$	$\delta = 1$	$\delta =$ 1.5
1.0	1	14.92 (6.71)	4.00 (5.31)	2.50 (3.84)	14.90 (6.70)	-0.60 (3.01)	-0.51 (1.69)
	3	14.59 (6.66)	2.14 (3.18)	1.24 (1.89)	15.03 (6.61)	-0.22 (1.24)	-0.18 (0.60)
	5	15.01 (6.65)	1.45 (2.45)	0.67 (1.34)	14.90 (6.61)	-0.18 (0.81)	-0.10 (0.39)
1.5	1	10.28 (6.24)	6.24 (6.00)	4.45 (5.12)	2.72 (6.20)	0.50 (3.85)	0.12 (2.33)
	3	10.12 (6.05)	3.66 (4.44)	2.27 (3.02)	2.72 (6.12)	0.31 (1.91)	0.03 (0.88)
	5	10.31 (6.08)	2.84 (3.73)	1.53 (2.35)	2.90 (6.02)	0.17 (1.23)	-0.01 (0.54)
3.0	1	10.95 (5.81)	10.03 (5.86)	8.30 (6.00)	5.01 (4.63)	3.48 (4.67)	2.34 (3.94)
	3	10.95 (5.60)	7.44 (5.91)	4.95 (4.97)	5.17 (4.44)	2.23 (3.45)	1.23 (2.24)
	5	11.31 (5.62)	6.18 (5.64)	3.88 (4.35)	5.20 (4.41)	1.66 (2.73)	0.80 (1.58)

Part (f). Performance of the Hawkins' estimator $\hat{\tau}_G$.

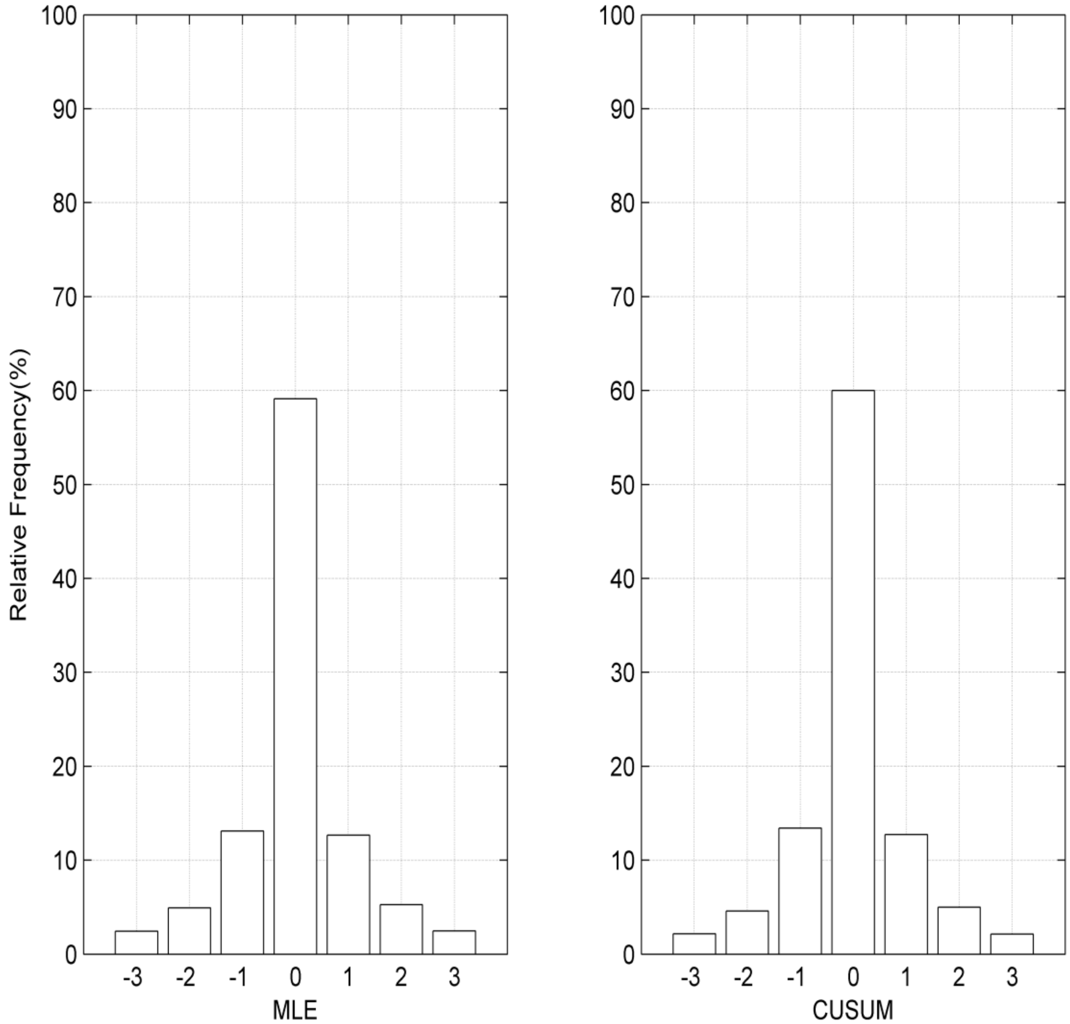
$G(0.5,1)$		τ/T =0.2			τ/T =0.5		
σ_1/σ_0	n	$\delta = 0.0$	$\delta = 1$	$\delta = 1.5$	$\delta = 0.0$	$\delta = 1$	$\delta =$ 1.5

1.0	1	14.91 (8.10)	8.79 (8.16)	8.30 (7.96)	14.99 (8.08)	0.40 (7.81)	0.37 (7.49)
	3	14.53 (8.65)	8.58 (8.48)	8.57 (8.10)	14.95 (8.50)	0.50 (8.37)	0.38 (8.14)
	5	14.93 (8.72)	9.05 (8.57)	8.81 (8.07)	14.94 (8.68)	0.30 (8.55)	0.47 (8.23)
1.5	1	8.52 (7.93)	8.07 (7.84)	7.91 (8.12)	0.13 (7.64)	0.15 (7.54)	-0.17 (7.40)
	3	7.17 (8.07)	6.98 (8.27)	6.56 (8.45)	0.25 (7.63)	-0.23 (7.43)	-1.16 (7.24)
	5	6.77 (8.02)	6.39 (8.18)	6.14 (8.44)	0.20 (7.16)	-0.66 (7.04)	-1.77 (6.79)
3.0	1	6.18 (7.50)	6.09 (6.82)	5.87 (6.76)	-0.01 (5.56)	0.87 (5.62)	0.74 (5.57)
	3	2.96 (5.85)	2.92 (5.42)	2.72 (5.38)	0.05 (3.92)	0.62 (3.89)	0.27 (3.82)
	5	1.83 (4.70)	1.85 (4.33)	1.58 (4.22)	0.09 (2.83)	0.41 (2.79)	0.16 (2.75)



Relative Frequency(%)
100
90
80
70
60
50
40
30
20
10
0
-3
-2
-1
0
1
2
3
MLE
CUSUM

Figure 1.



100
90
80
70
60
50
40
30
20
10
0
Relative Frequency(%)
-3 -2 -1 0 1 2 3
MLE
CUSUM

Figure 2.

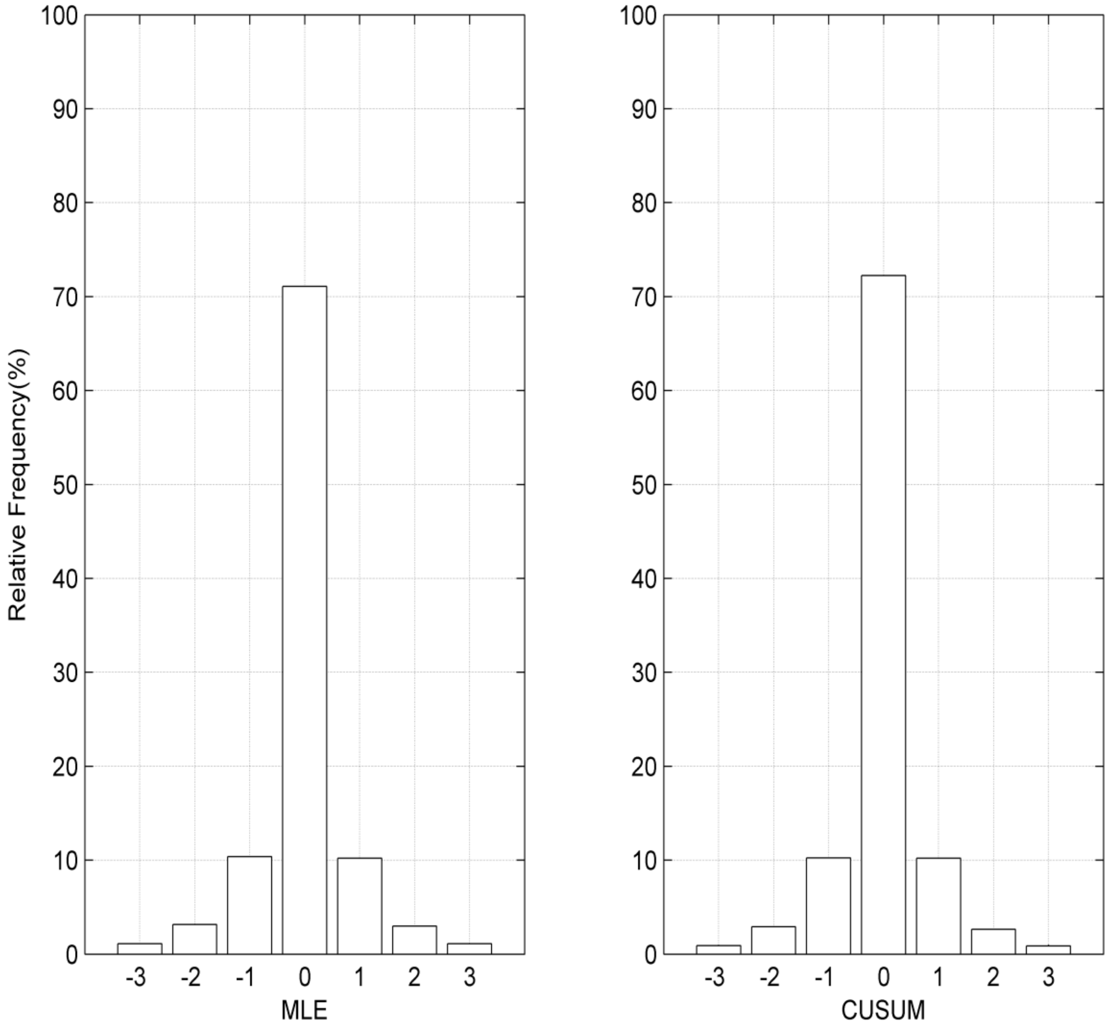

100
90
80
70
60
50
40
30
20
10
0
Relative Frequency(%)
-3
-2
-1
0
1
2
3
MLE
CUSUM

Figure 3.

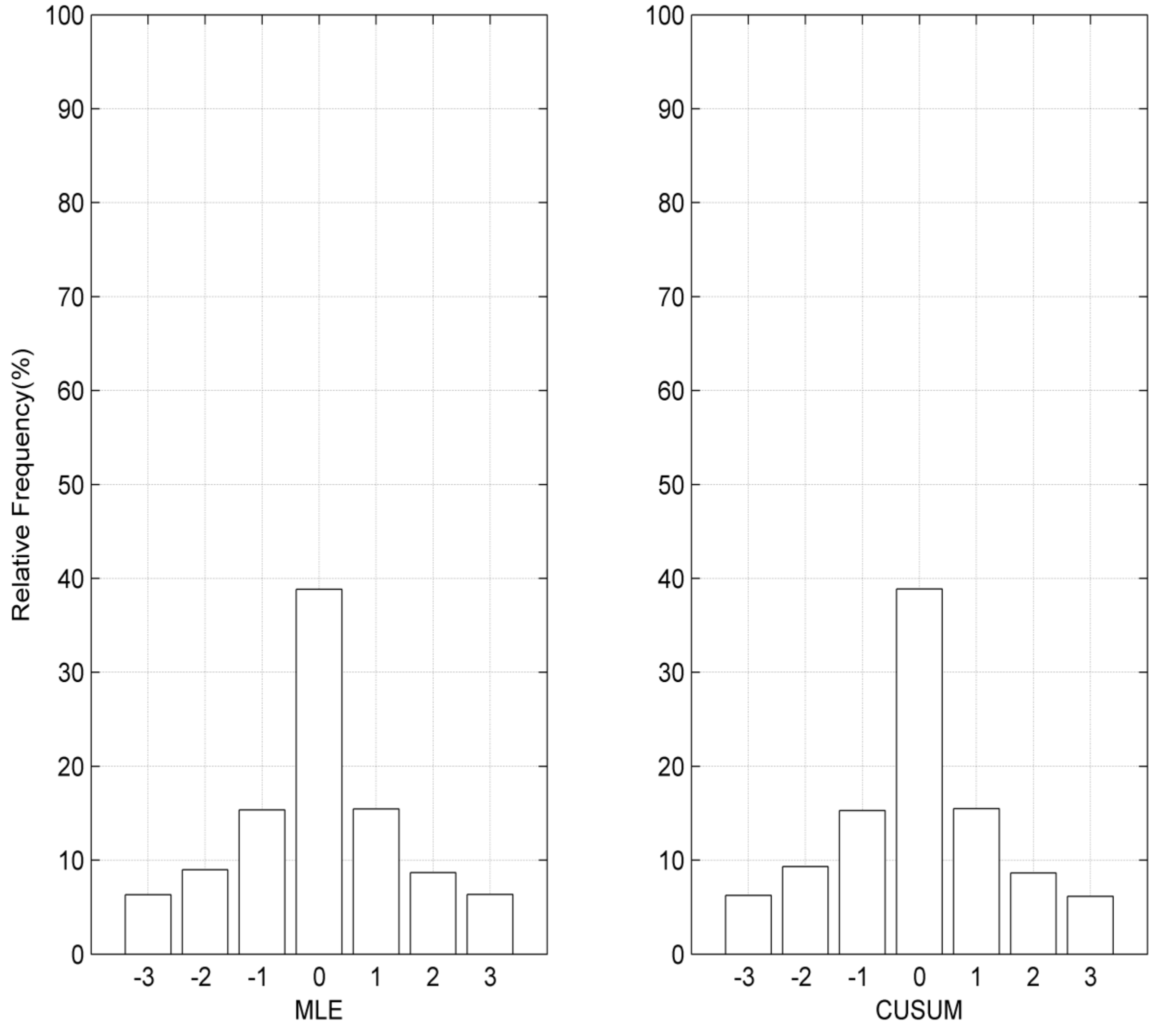

Relative Frequency(%)
100
90
80
70
60
50
40
30
20
10
0
-3
-2
-1
0
1
2
3
MLE
CUSUM

Figure 4.

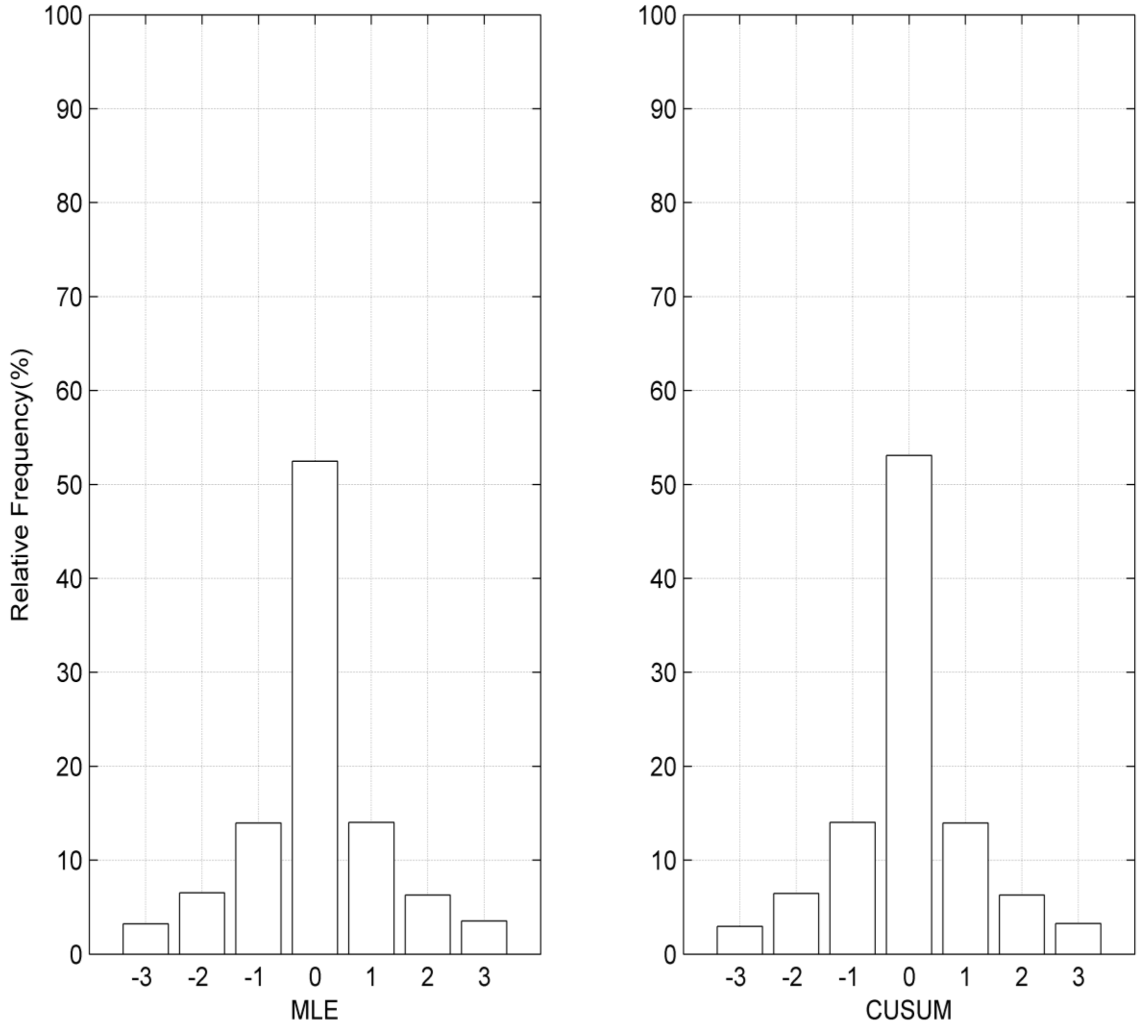

100
90
80
70
60
50
40
30
20
10
0
Relative Frequency(%)
-3 -2 -1 0 1 2 3
MLE
100
90
80
70
60
50
40
30
20
10
0
-3 -2 -1 0 1 2 3
CUSUM

Figure 5.

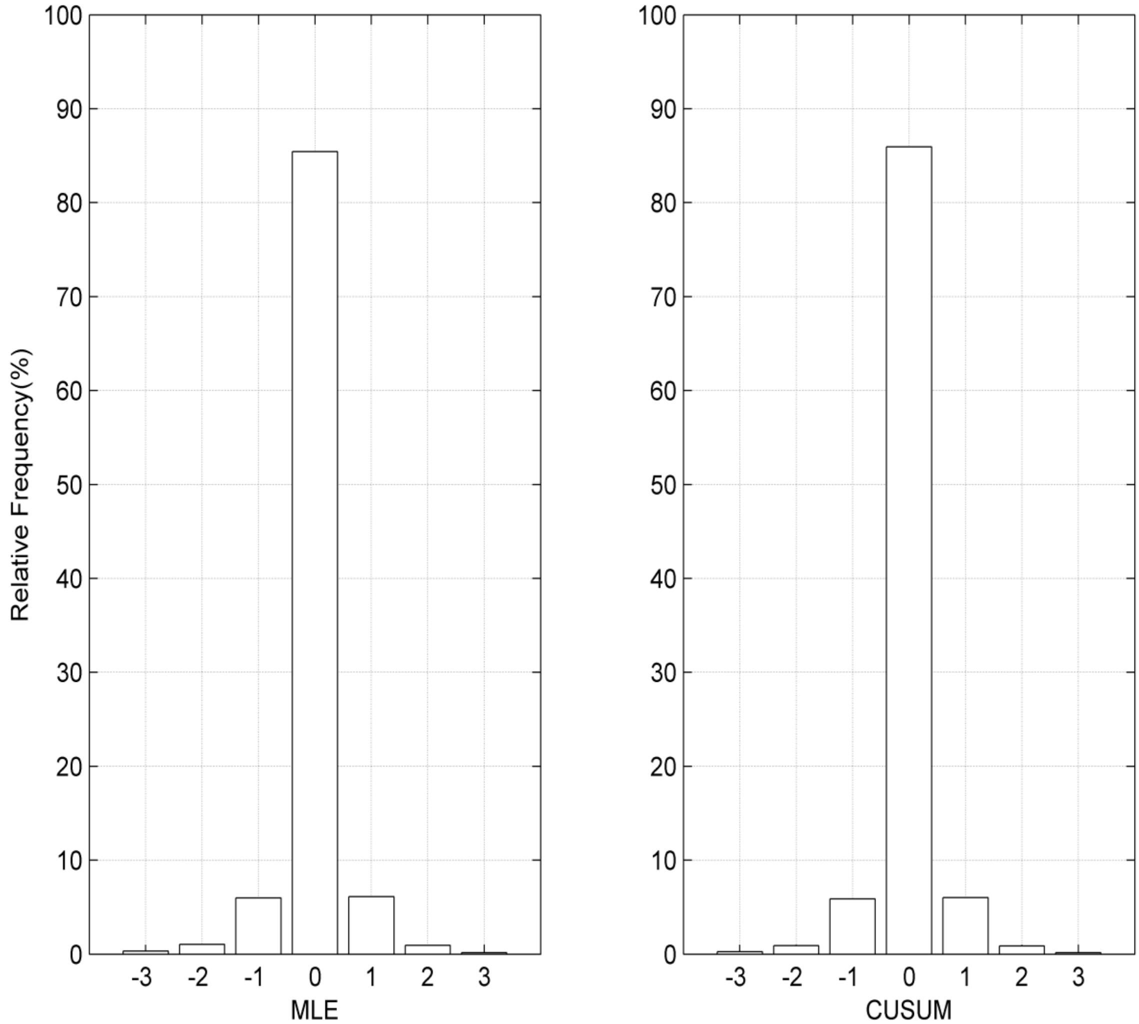

Relative Frequency(%)
100
90
80
70
60
50
40
30
20
10
0
-3 -2 -1 0 1 2 3
MLE
CUSUM

Figure 6.

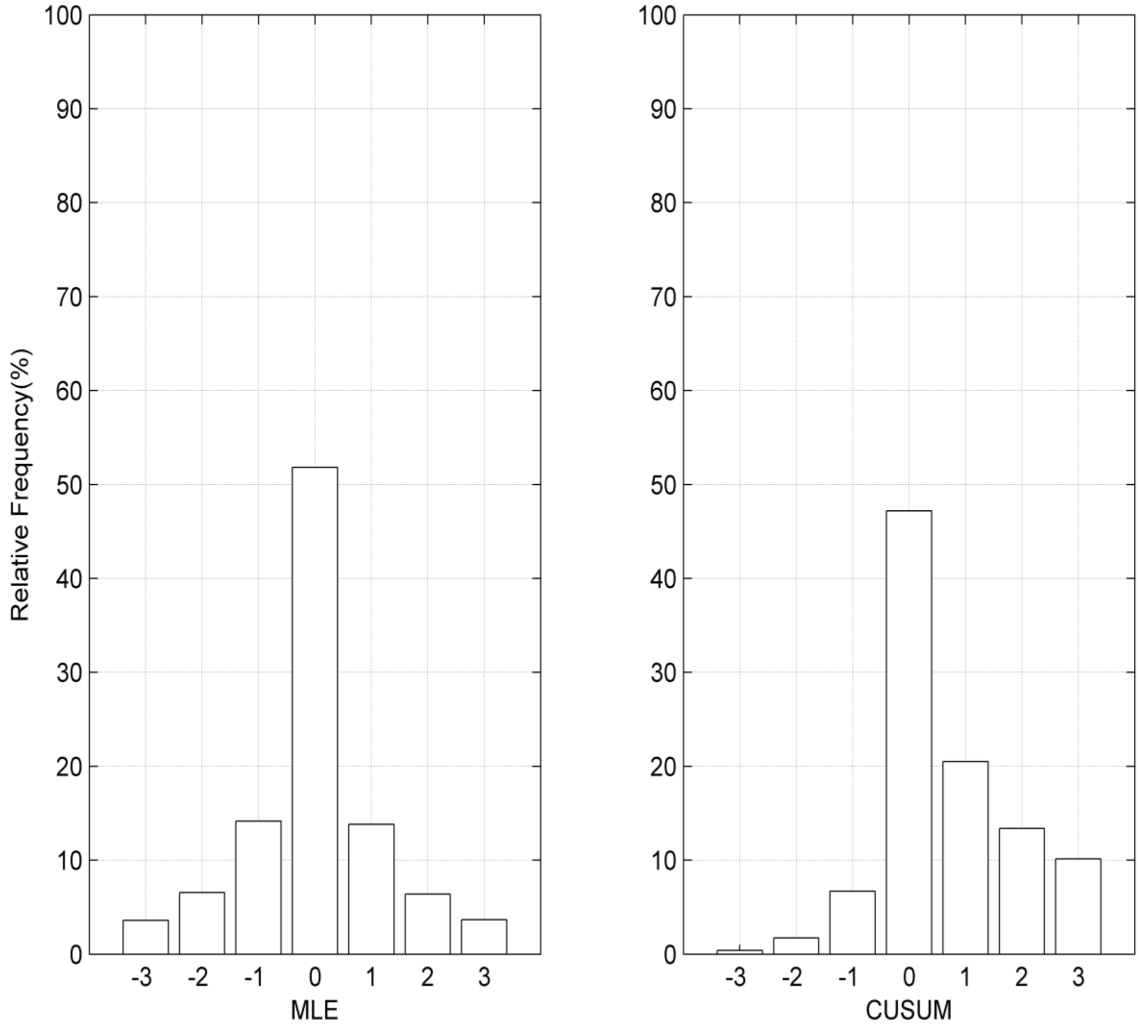

Relative Frequency(%)
100
90
80
70
60
50
40
30
20
10
0
-3 -2 -1 0 1 2 3
MLE
CUSUM

Figure 7.

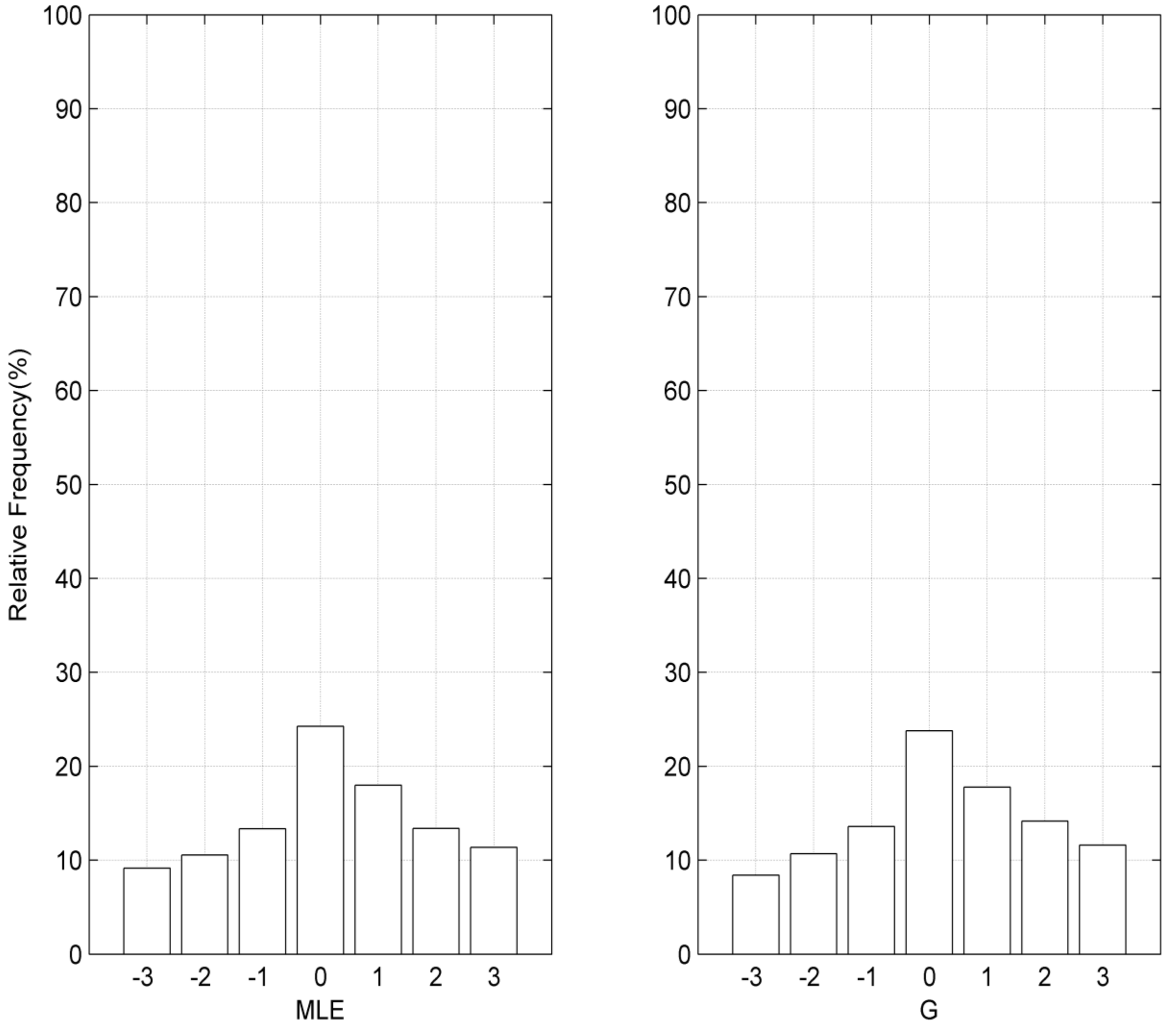

100
90
80
70
60
50
40
30
20
10
0
Relative Frequency(%)
-3 -2 -1 0 1 2 3
MLE
100
90
80
70
60
50
40
30
20
10
0
-3 -2 -1 0 1 2 3
G

Figure 8.

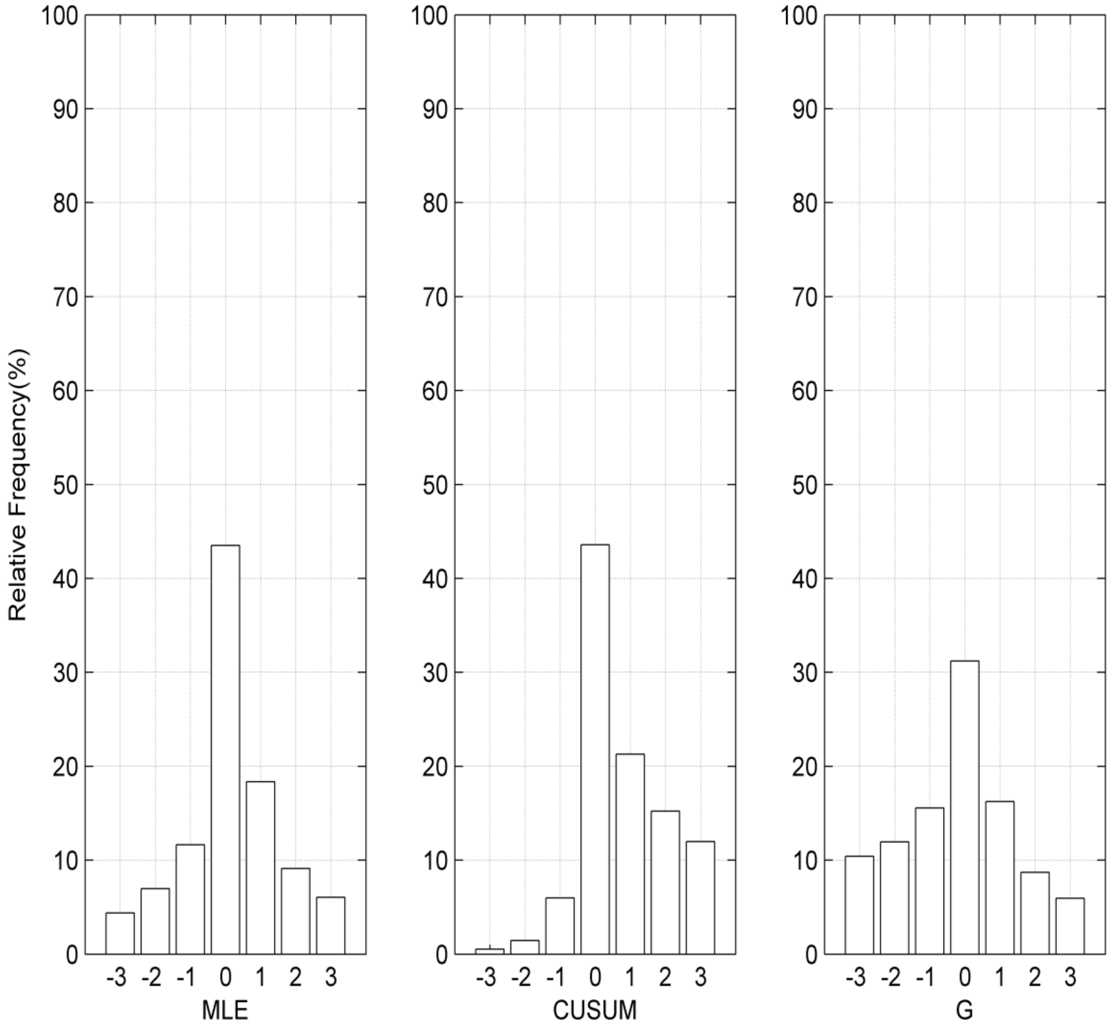

Relative Frequency(%)
100
90
80
70
60
50
40
30
20
10
0
-3 -2 -1 0 1 2 3
MLE
CUSUM
G

Figure 9.

 ACCEPTED MANUSCRIPT